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Master's Thesis

Asymmetric Influence and Interventions in Climate Change Belief Systems

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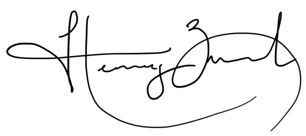
Computational Science

Declaration of Authorship

I, Henry Zwart, declare that this thesis, entitled 'Asymmetric Influence and Interventions in
Climate Change Belief Systems' and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at the University of Amsterdam.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed:



Date: 08 August 2026

UNIVERSITY OF AMSTERDAM

Abstract

Faculty of Science
Informatics Institute

Master of Science in Computational Science

Asymmetric Influence and Interventions in Climate Change Belief Systems

by Henry ZWART

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Acknowledgements

Thank the people that have helped: supervisors, family, etc.

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Abbreviations

CSL **C**omputational **S**cience **L**ab

UvA **U**niversiteit **v**an **A**msterdam

Chapter 1

Introduction

Our subjective interpretations of the world, natural phenomena, and those around us are dependent on a collection of beliefs about how things are, and how they work. These beliefs and attitudes are highly interdependent, related by logical and psychological associations in what are often referred to as belief systems (Converse, 1964; Fishbein & Ajzen, 1977). Beliefs and attitudes are also subject to social dynamics (Galesic et al., 2021), as evidenced by observed geographic segregation of political attitudes in the United States (Brown & Enos, 2021). Moreover, beliefs and attitudes held by an individual have external consequences for behavioural decisions (Granovetter, 1978) and are in turn influenced by these decisions (Fishbein & Ajzen, 1977; Olson & Stone, 2005), often in the form of reinforcing feedback loops. Put plainly, the beliefs and attitudes which allow us to make internal sense of the world inevitably shape our collective impact on it.

Distinct beliefs and attitudes behave differently within a belief system. Some attitudes, such as political ideology (e.g., conservativism or liberalism) are highly stable over time (Green & Platzman, 2024; Kiley & Vaisey, 2020; Osborne & Sibley, 2020). Others appear to be more malleable (e.g., attitudes toward particular political candidates). In addition, some beliefs and attitudes appear to be more influential than others. For instance, one might intuitively expect political alignment to be dependent on individuals' policy preferences. However, Unsworth & Fielding (2014) demonstrate that causal influence may actually flow primarily in the opposite direction. When shown identical policies with different partisan framings, individuals' support for those policies shifts in line with their own political alignment, suggesting that partisan identity may shape policy preferences more so than in the other direction.

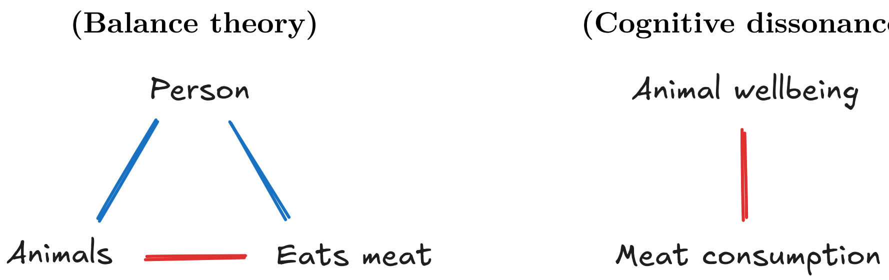
Given belief systems' interdependent nature, understanding the behaviour of any single belief (or attitude) requires consideration not only of how it interacts with other beliefs, but also how *those beliefs* interact with the broader belief system. This has motivated empirical approaches to studying belief system structure based on network science, which treat belief systems as undirected networks describing the pairwise correlational structure (edges) between distinct beliefs and attitudes (nodes) (Boutyline & Vaisey, 2017; Costantini et al., 2015; Dalege et al., 2017; Epskamp et al., 2012; Epskamp, Borsboom, et al., 2018).

Analysis of *belief system networks* using network science methods has proven valuable for understanding belief system structure and the roles of specific beliefs and attitudes in topic-specific belief systems. Using cross-national surveys, several studies have examined geographic variation in belief system structure on topics including politics (Keskintürk, 2022; van Noord et al., 2025) and climate change (Lee et al., 2025), comparing both specific relations and whole-network features such as density (the proportion of realised connections), and inconsistency (the number of negative correlations). Similarly, Chambon et al. (2023) examines how beliefs and attitudes relating to COVID-19 changed over time in the Netherlands during the start of the global pandemic. Despite ongoing debate about its usefulness and applicability to belief system networks (cf. Bringmann et al. (2019)), node centrality is frequently used to assess belief position within a belief system (Borsboom et al., 2021; Brandt et al., 2019), relative influence (Robinaugh et al., 2016), or structural importance (Dalege et al., 2017; Fonseca-Pedrero, 2018; Hevey, 2018). Chambon et al. (2022) demonstrate that node centrality may also contribute to the propagation of belief-level interventions, in the context of a belief system on COVID-19.

Although belief system networks characterise the statistical associations between beliefs and attitudes in a belief system, they make no claims about the nature of these associations, how they arise, nor their implications for belief dynamics. Several explanatory models have been proposed to address these shortcomings, generally considering endogenous belief system dynamics as driven by individuals' efforts to achieve consistent sets of beliefs and attitudes; that is, which are perceived as mutually compatible. This perspective is particularly amenible to formalisation under the framework of complex systems, treating belief system dynamics as arising from micro-level attempts to resolve perceived inconsistencies between beliefs and attitudes.

Current theories (and models) of belief system dynamics have been especially influenced by two classical theories of belief and attitude consistency: Heider's *balance theory* (1946) and Festinger's *theory of cognitive dissonance* (1962). Heider's balance theory interprets attitudes as signed, affective associations between concepts. Inconsistency arises when three concepts are related in a triangle for which the product of edge weights is negative. Festinger's theory of cognitive dissonance, on the other hand, takes the view that beliefs (and by extension, attitudes) can be more or less compatible with one another.

The difference between these theories is best clarified with an example (illustrated below). Consider a person who cares deeply about animal wellbeing but who also eats meat. Heider (*left*) considers the cognitive inconsistency as arising from the negative association between animals and meat consumption (interpreted as 'harm'). To resolve this, the individual could adjust their attitudes toward animal wellbeing, change their dietary habits, or limiting the perceived association between animals and meat consumption. Festinger (*right*) relates the individual's attitudes directly rather than via their conceptual referrents, but arrives at the same conclusion, that the person will tend to change one of the two attitudes, or reduce their perceived negative association.



Balance theory and the theory of cognitive dissonance have given way to two families of belief system models basd on the statistical physics notion of energy minimisation (Ising, 1925). The present study primarily considers models inspired by Festinger's theory of cognitive dissonance, which is more so concerned with relations *between* beliefs and attitudes than attitudes as relating concepts. For models based on Heider's balance theory, we refer the reader to Greenwald et al. (2002), Rodriguez et al. (2016), and Aiyappa et al. (2024).

The *Causal Attitude Network* (CAN) model (Dalege et al., 2016) formalises Festinger's theory of cognitive dissonance in an Ising-style model, also drawing on the more recently-proposed connectionist perspective on attitude change (Monroe & Read, 2008). It can be viewed as a theory-driven variant of the belief network approach described above. While the Ising model was originally presented as a model of ferromagnetic behaviour (Ising, 1925), it has since been variously applied to describe diverse systems of interacting variables which attract or repel one another (Nguyen et al., 2017). In the CAN model, beliefs and attituds are represented as spins which take on values in the set $\{-1, +1\}$, representing two opposing states (e.g., 'climate change is happening' versus 'climate change is *not* happening' or 'support' versus 'oppose'). These are related by reinforcing (+) or opposing (−) interaction effects, based on the notion of cognitive dissonance. The model dynamics are described in terms of energy (read 'inconsistency') minimisation. That is, individuals tend to reduce cognitive dissonance, such that positively associated beliefs generally agree, and negatively associated beliefs generally disagree.

Importantly, the CAN model defines interaction relations as symmetric, such that related beliefs exert equal reinforcing or opposing influence on one another. Under this assumption the underlying Ising model satisfies detailed balance and can be considered an equilbrium system (Christensen & Moloney, 2005). This is mathematically convenient, as it allows for implicit definition of the CAN model dynamics using the Boltzmann distribution, such that each belief system state is observed with constant probability as a decreasing function of that state's inconsistency (i.e., energy). Highly consistent states are observed frequently, while highly inconsistent states are rarely observed.

In spite of its simple foundations, the CAN model and recent variants have proven effective at capturing observed phenomena and demonstrated value for predicting and testing hypotheses about belief and attitude change. For instance, both Dalege et al. (2018) and van der Maas et al. (2020) show that such models may explain how merely thinking about a topic can induce more extreme views. In several cases the CAN model (or analogous models) have been used to explore the potential effects of intervention or persuasion attempts on belief system dynamics via simulation (Bertero & Franetovic, 2025; Dalege et al., 2017; Lunansky et al., 2022; Schlicht-Schmälzle et al., 2018). Furthermore, van der Maas et al. (2020) and Dalege et al. (2025) demonstrate that extensions of the CAN model which incorporate social influences on belief and attitude dynamics are also highly expressive. In both cases, the authors model social network influences via theoretically simple social cohesion mechanisms (i.e., the assumption that individuals who interact tend to align their beliefs and attitudes). This turns out to be sufficient to capture various phenomena observed in reality, including: (i) attitude polarisation, (ii) how interventions can drive extreme attitudes on topics that has previously received limited attention, and (iii) how minority views propagate through a population (Dalege et al., 2025).

However, although mathematically convenient, symmetric interactions are not psychologically necessary. Indeed, this has been broadly acknowledged within the belief systems modelling literature as a limitation of current approaches. Most authors explicitly mention either causal directionality underlying inferred bi-directional associations, or asymmetric/non-reciprocal influences as *plausible* (Brandt, 2022; Brandt & Sleegers, 2021; Epskamp, van Borkulo, et al., 2018; Keskintürk, 2022; Lee et al., 2024; Monroe & Read, 2008; Powell et al., 2023). van der Maas et al. (2020) go so far as to assert that asymmetric influences are *likely*, for instance, in relations linking attitudes and behaviour. Moreover, there exists some empirical evidence supporting the existence of asymmetric interactions. For instance, Chambon et al. (2023) observe directional differences in temporal networks (van der Krieke et al., 2015) used to study belief systems relating to COVID-19; however, as identifying asymmetry was not a core aim of this study they do not test for the significance of these differences.

Despite general agreement on the plausibility of asymmetric interactions, current research remains predominantly focused on symmetric models[1]. A handful of authors make reference to simulation studies in which asymmetric edges are used and found to have minimal impact on model dynamics (Monroe & Read, 2008; van der Maas et al., 2020). However, these experiments are rarely described in detail, and we are not aware of any instances in which asymmetric models are estimated from data. For the purposes of mean-field approximation, van der Maas et al. (2026) argue that theoretical results derived for Ising-style belief system models are robust to asymmetric and non-reciprocity assumptions. However, they note that studies outside the belief system modelling literature demonstrate that non-reciprocal Ising models can exhibit considerably more complex behaviour than the symmetric model (cf. Avni et al. (2025)).

One area in which the effects of asymmetric influence are likely to surface is the assessment of belief importance, often quantified using node centrality indices (as discussed above). While most undirected-network centrality indices have directed-network analogues, these can differ substantially in value (Bringmann et al., 2019). This becomes especially critical when belief centrality is used to predict intervention effectiveness. For instance, beliefs with many connections (i.e., high strength or degree centrality) are often considered both (i) good targets for interventions intended to propagate to other beliefs (cf. Chambon et al. (2022)), and (ii) more resistant to such interventions, on account of their received influence from adjacent beliefs (cf. Brandt & Vallabha (2023)). However, both conclusions rely on the assumption of bi-directional influence. When influence may be asymmetric or non-reciprocal, we may find that some beliefs are characterised almost exclusively by incoming or outgoing interactions, leading to situations where well-connected beliefs have limited influence on other beliefs or little-to-no resistance to interventions, respectively.

At the time of writing, empirical and theoretical studies into the existence, nature, and potential structure of asymmetric influences among beliefs remain scarce. Of particular

[1]Bayesian network models are a notable exception, which consider belief systems as directed acyclic graphs (Cook & Lewandowsky, 2016; Powell et al., 2023). However, these do not allow for reinforcing relationships. Moreover, they fall victim to the same problem as faced by the statistical belief system networks—while they describe the relationships observed in data, they provide no explanation for the nature of these relationships.

concern, the absence of theoretical models of asymmetric influences appears to be self-perpetuating, with some studies citing historical trends as justification for continued focus on symmetric models (Brandt, 2022, p. 3). Consequentially, it remains unclear as to whether asymmetric influence—should it exist—meaningfully impacts belief system dynamics, or the conclusions we draw from models thereof.

In this study, we investigate the prevalence of asymmetric influence among beliefs and attitudes about climate change in the US, as well as the dynamic implications of symmetric and asymmetric modelling assumptions for belief-level interventions. Using a combination of data-driven and simulation-based methods, we will address the following four research questions:

RQ1. To what extent are belief-level influences *symmetric* or *asymmetric*, in models of climate change belief systems inferred from the climate beliefs dataset?

RQ2. How do asymmetric and symmetric beliefs systems differ with regards to intervention strategy and effectiveness, in models inferred from the climate beliefs dataset?

RQ3. How do intervention outcome and effectiveness vary between individuals with different initial conditions in asymmetric belief systems inferred from the climate beliefs dataset?

RQ4. How do asymmetric belief systems inferred from the climate beliefs dataset vary between conservative and liberal individuals?

This study is divided into two halves. In the first half we establish a framework for modelling belief systems with asymmetric influence relations, and estimating such models from time-series data. In the second half, we then use this framework to address the research questions listed above.

In Chapter 2 we introduce the **Kinetic Belief System** model (KBS), a kinetic Ising model formulation (Fredrickson & Andersen, 1984; Glauber, 1963) of the Causal Attitude Network (CAN) model (Dalege et al., 2016). KBS represents directed belief influences using distinct parameters, so is well-suited for studying asymmetry in belief systems. KBS' dynamics are explicitly time-dependent, enabling straightforward analysis of belief system dynamics, including post-intervention behaviour, on an *individual* basis. This contrasts past studies that analyse intervention effects via simulation on the CAN model (Bertero & Franetovic, 2025; Dalege et al., 2017; Lunansky et al., 2022; Schlicht-Schmälzle et al., 2018) or GGM models (Wu et al., 2026), which do consider neither individuals' pre-intervention belief states, nor the time-scale of model dynamics. Chapter 3 then outlines a parameter estimation method for KBS based on maximum likelihood estimation. The proposed method uses knowledge of a pre-defined soft binarisation function to robustly estimate binary model parameters from survey data that is not necessarily binary, without requiring explicit binarisation. We use this method to calibrate the KBS model to a two-wave longitudinal dataset comprising beliefs and attitudes relating to climate change (see below).

We then subsequently use the calibrated model to address our research questions in the second half of the study. Inspired by approaches used in the context of the CAN model, in Chapter 3 we formalise interventions in Ising-style belief system models as additional influencing variables. We also give algorithms for: (i) quantifying expected intervention effects on an individual basis and (ii) comparing effects between symmetric and asymmetric models, using Common Random Numbers to ensure result comparability. We assess the existency of asymmetry and its population-level implications for intervention dynamics in Chapter 5, and heterogeneity in intervention outcomes and belief systems in Chapter 6, relating these back to our research questions in Chapter 7, also discussing broader implications for belief system modelling.

The data used for this study is sourced from the Longitudinal Panel of Perceptions About Climate Change and Covid representative longitudinal survey (**CCCV**), collected in the US between 2020 and 2023 (Constantino et al., 2022). Comprehensively validating this dataset and constructing the targeted dataset of beliefs and attitudes relating to climate change used in our experiments constituted substantial components of this investigation, and contribute to future use of the CCCV dataset. We detail both processes in Chapter 9.

In its totality, this study presents a theory-driven approach to studying the structure and dynamics of asymmetric belief systems. We apply this to investigate the existence and prevalence of asymmetry in an observational context regarding beliefs and attitudes about climate change, and the consequences of symmetry and asymmetry assumptions for reasoning about intervention dynamics.

Chapter 2

Non-equilibrium belief systems

TODO: "The DT-VAR model suffers from the problem of time-interval dependency (Gollob & Reichardt, 1987)" (Ryan & Hamaker, 2022)

- Ours does not(?) because we use temporal data, and include self-interaction effects which set timescale.

Our theory of belief system dynamics rests on three main assumptions. First, we assume that beliefs and attitudes can be represented as discrete random variables. This allows us to consider beliefs and attitudes as entities characterised by an instantaneous state. In particular, this assumption contrasts alternative views that beliefs and attitudes are dispositional or the result of interactions with the environment, with no associated state of their own (**CITATIONS**).

Second, we assume that the state distribution for any given belief or attitude is conditional on the previous states of all others, formalising the idea that past beliefs and attitudes influence present ones. Finally, we assume that these conditional distributions are time-invariant. While belief and attitude *states* may change over time, the dynamics by which this happens do not.

TODO: Make clear that when we say attitude we mean evaluative state, not precise definition used in (Dalege et al., 2016).

> **Important**
>
> The terms **belief** and **attitude** are presently ambiguous. They can refer to generic concepts (e.g. belief regarding the contents of a box, or attitude toward …) or specific instances of those concepts (e.g., 'I believe that the box is empty', or '…').

> For the remainder of this thesis, unless stated otherwise, we adopt the *generic* sense. That is to say that we use the term 'belief' without assuming any particular epistemic state, and 'attitudes' without assuming any particular disposition.
>
> For simplicity, we will also typically use the term **belief** to refer to both beliefs and attitudes, except when the distinction is important.

Consider a set of belief $\boldsymbol{S} = \{S_1, S_2, ..., S_N\}$, where for each $i \in 1..N$, the belief which takes values from . Given the above assumptions, we can describe the trajectory of a belief or attitude S_i with domain Ω_i as a sequence of states

$$\{s_i^t\}_{t=1}^{\infty} \quad \text{where} \quad s_i^t \in \Omega_i \tag{2.1}$$

characterised by the family of conditional distributions $P\left(\sigma_i^{t+1} \mid \boldsymbol{\sigma}^t\right)$ for $1 < t \in \mathbb{N}$ and an initial state $P(\sigma_i^1)$.

Noting that the state of a belief S_i depends only on the previous states of all other beliefs and attitudes, it follows that any pair of beliefs or attitudes S_i and S_j are conditionally independent at a given timestep given all previous belief and attitude states. In particular, this means we can describe the evolution of the complete set of beliefs and attitudes as a Markov process:

$$\{\boldsymbol{s}^t\}_{t=1}^{\infty} \quad \text{where } \boldsymbol{s}^t \in \Omega_1 \times \cdots \times \Omega_N \tag{2.2}$$

with transition probability $P(\boldsymbol{\sigma}^{t+1} \mid \boldsymbol{\sigma}^t) = \prod_i^N P\left(\sigma_i^{t+1} \mid \boldsymbol{\sigma}^t\right)$ for $1 < t \in \mathbb{N}$ and initial state probability $P(\boldsymbol{\sigma}^1) = \prod_{i=1}^N P(\sigma_i^1)$.

We use the term **belief system** to refer to the combination of a collection of attitudes $\boldsymbol{S}$ and its associated transition probability distribution $P\left(\boldsymbol{S}^{t+1} \mid \boldsymbol{S}^t\right)$. In this conceptualisation, the task of modelling a belief system reduces to describing how the instantaneous configuration of belief and attitude states affects the distribution over possible future states.

2.1 Non-equilibrium belief system model

TODO:

- Builds on the CAN model, also known as the 'Ising model of attitudes' (van der Maas et al., 2020)
  - Also (Brandt & Sleegers, 2021)

- Points of contrast:
  - Asymmetric interactions
  - Self-interactions
- (Brandt & Sleegers, 2021) also uses transition probabilities like ours
  - Also (Haslbeck et al., 2021)

NOTE: Self-interaction effects:

- (Brandt & Sleegers, 2021, p. 22): "we made the simplifying assumption that a node does not affect itself".

When studying the *symmetric* Ising model it is common to assume that the model is at equilibrium, such that the transition probability is stationary and given by the Boltzmann distribution (Cardy, 1996; Christensen & Moloney, 2005).

For the purposes of this study, however, we are interested in intervention dynamics, which are inherently non-equilibrium. To see why this is the case, notice that the purpose of intervention is to change the distribution of observed states. This is true both interventions intended to change the dominant state (e.g., to promote a sustainable alternative to a typical unsustainable behaviour) or reinforce it. Any belief system model intended for studying intervention dynamics therefore cannot assume equilibrium, and cannot define model dynamics using the Boltzmann distribution.

We instead define the conditional distribution $P(\boldsymbol{S}^{t+1} = \boldsymbol{s} \mid \boldsymbol{S}^t)$ explicitly. Recall that the states of each pair of spins S_i, S_j at time t are conditionally independent random variables, given knowledge of the previous configuration $\boldsymbol{S}^t$. It therefore suffices for us to describe the distribution over spin states for an individual spin S_i at time $t+1$, as conditional on $\boldsymbol{S}^t$.

As in the standard Ising model formulation, we will assume that the model typically evolves in the direction of lower energy; however, rather than defining the energy at the level of the model using the Hamilonian, we consider the energy experienced by S_i as a result of its baseline activation (also known as a *local field effect*), and interactions with other spins.

The baseline activation $h_i \in \mathbb{R}$ describes the tendency for S_i to take on the value $+1$ in the absence of interactions with other spins, or, more generally, when pairwise spin interactions have a net-effect of zero. S_i tends to adopt the value $+1$ under these circumstances, iff, $h_i > 0$, and -1 iff $h_i < 0$. This tendency increases with the magnitude of h_i.

Other spins influence S_i's state through alignment or opposition relations. For a spin S_j, we define an interaction effect J_{ji} (read *'influence of j on i'*). When J_{ij} is positive or negative, S_i is influenced to adopt the state S_j^t or $-S_j^t$ respectively. In the special case when $j = i$, we refer to J_{ii} as a *self-influence* of *self-interaction* effect. Positive self-influence effects reflect the inertia of S_i, i.e., the tendency to sustain a particular belief or attitude irrespective of other beliefs and attitudes. Negative self-influence effects are not clearly interpretable. In addition to modelling differences in inertia, the inclusion of self-interaction effects allows us to capture the timescale at which the system, as a whole, evolves.

The energy experienced by S_i for a particular spin state $s \in \{-1, +1\}$ is then the result of s in combination with the baseline activation and influence effects from all spins with edges to S_i:

$$H_i(s \mid \boldsymbol{s}^t) = h_i \cdot s + \sum_j A_{ji} J_{ji} s_j^t \cdot s \tag{2.3}$$

where $\boldsymbol{A} \in \{0,1\}^{n \times n}$ is a pairwise adjacency matrix, with $A_{ji} = 1$, iff, S_j influences S_i. Dividing the common factor of s, we can equivalently write Equation 2.3 as

$$H_i(s \mid \boldsymbol{s}^t) = s \cdot h_i^{\text{eff}}(\boldsymbol{s}^t) \tag{2.4}$$

where $h_i^{\text{eff}}(\boldsymbol{s}^t) = h_i + \sum_j A_{ji} J_{ji} s_j^t$ is the effective baseline activation experienced by S_i at time t. We then define the probability that S_i adopts the state s at time t as:

$$\begin{aligned} P\big(S_i^{t+1} = s \mid \boldsymbol{S}^t = \boldsymbol{s}\big) &= \frac{e^{\frac{1}{T} H_i(s \mid \boldsymbol{s})}}{\sum_{s' \in \{0,1\}} e^{\frac{1}{T} H_i(s' \mid \boldsymbol{s})}} \\ &= \frac{e^{\frac{1}{T} H_i(s \mid \boldsymbol{s})}}{e^{\frac{1}{T} H_i(s \mid \boldsymbol{s})} + e^{-\frac{1}{T} H_i(-s \mid \boldsymbol{s})}} \\ &= \frac{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\boldsymbol{s})}}{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\boldsymbol{s})} + e^{-\frac{1}{T} s \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \end{aligned} \tag{2.5}$$

For $s = +1$, this reduces to the logistic function:

$$\begin{aligned} p_i^{\boldsymbol{s}} := P\big(S_i^{t+1} = +1 \mid \boldsymbol{S}^t = \boldsymbol{s}\big) &= \frac{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}}{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})} + e^{-\frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \\ &= \frac{1}{1 + e^{-2 \cdot \frac{1}{T} \cdot h_i^{\text{eff}}(\boldsymbol{s})}} \\ &= \text{logistic}\big(2 h_i^{\text{eff}}(\boldsymbol{s}) / T\big) \end{aligned} \tag{2.6}$$

Therefore the complete conditional distribution is given by:

$$\begin{aligned} P(\boldsymbol{S}^{t+1} = \boldsymbol{s}^{t+1} \mid \boldsymbol{S}^t = \boldsymbol{s}^t) &= \prod_{i=1}^{n} P(S_i^{t+1} = s_i \mid \boldsymbol{S}^t = \boldsymbol{s}^t) \\ &= \prod_{i=1}^{n} \Big[\Big(\frac{1+s_i}{2}\Big) p_i^{\boldsymbol{s}^t} + \Big(\frac{1-s_i}{2}\Big)\big(1 - p_i^{\boldsymbol{s}^t}\big) \Big] \end{aligned} \tag{2.7}$$

Where the initial distribution $P(\boldsymbol{S}^0)$ is specified explicitly.

2.2 Symmetric and asymmetric belief systems

Let $\mathcal{M}$ be a belief system model defined as in the preceding section, comprising $N \in \mathbb{N}$ beliefs or attitudes, with adjacency matrix $\boldsymbol{A} \in \{0,1\}^{N \times N}$, interaction effect matrix $\boldsymbol{J} \in \mathbb{R}^{N \times N}$, and baseline activations $\boldsymbol{h} \in \mathbb{R}^N$.

When each entry in $\boldsymbol{J}$ is independent of all others, we say that $\mathcal{M}$ is an **asymmetric belief system**. The term *asymmetric* here refers to the directed relations between a pair of nodes. In an asymmetric belief system, for any pair of distinct nodes $S_i \neq S_j$, it may be the case that an influence relation exists only in one direction, or that the directed relations differ in magnitude.

In the special case where we constrain $A_{ij} = A_{ji}$ and $J_{ij} = J_{ji}$ for all $i, j \in [1, N]$, we instead say that $\mathcal{M}$ is a **symmetric belief system**. In a symmetric belief system all interactions are directionally-equivalent. This can be considered a simple extension of the symmetric Ising model with (i) self-interaction terms and (ii) temporal dynamics, thus the symmetric belief system model tends toward an equilibrium steady state.

2.3 Simulation via Glauber dynamics

It is straightforward to draw samples from a belief system model using Glauber dynamics, given the conditional probability distribution defined in Equation 2.7.

Given an initial state $\boldsymbol{s}^0 \in \{-1,+1\}^N$, we sample a sequence of $T \in \mathbb{N}$ subsequent configurations:

$$\{\boldsymbol{s}^t\}_{t=0}^{T}, \quad \text{where each } \boldsymbol{s}^{t+1} \sim P(\boldsymbol{S}^{t+1} \mid \boldsymbol{S}^t) \tag{2.8}$$

Each belief or attitude has the opportunity to update during every time interval. This update routine is referred to as *synchronous* Glauber dynamics, contrasting *asynchronous* Glauber dynamics, in which only one spin can update during a given interval.

2.4 Modelling interventions

TODO:

- Discuss (Dalege et al., 2025), which is somewhat analogous.

  In the NB model exogenous changes (other peoples' beliefs) affect your belief system via an interaction term to the belief about that state. Our approach is analogous. The exogenous change affects our belief (say, about the state of climate change) via an interaction term.

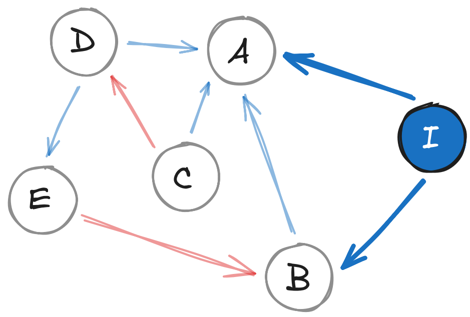
Other intervention studies:

- Simulation-based approaches that mirror ours:
  - (Dalege et al., 2017)
  - (Schlicht-Schmälzle et al., 2018)
  - (Lunansky et al., 2022)
  - (Bertero & Franetovic, 2025)
  - (Wu et al., 2026)

  All use symmetric models. AFAIK none use glauber dynamics, i.e., assume equilibrium right after intervention. So don't account for temporal dynamics, belief stability, individual effects, etc.
- Control theory approach (Henry et al., 2022)
- Continuous-time approach (avoids the discrete time problem) (Ryan & Hamaker, 2022)

We now outline our approach to modelling interventions in the asymmetric belief system model. In this study we consider interventions which affect the *state* of a belief system. Notably, this excludes interventions intended to influence the structure of a belief system, for instance, by changing the existence of effect size of influence relations between beliefs and attitudes.

Let $\mathcal{M}$ be a belief system model with parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$. We can consider an intervention as an auxiliary node, I, in the belief system network, with state fixed at a particular value and outgoing edges toward a subset of beliefs and attitudes:



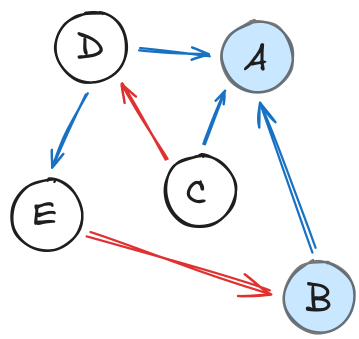
The intervention node I exerts influence on the belief system nodes A and B. Let us consider the effective baseline activation at node A and time $t+1$, as defined in Equation 2.3, given a previous configuration $\boldsymbol{s}^t$:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = h_A + \sum_{\text{node } j} A_{jA} J_{jA} s_j^t \tag{2.9}$$

Since the state of I is fixed (in this case, at $+1$), the contribution of the intervention node to A's effective activation baseline is constant. We can then re-write Equation 2.9, interpreting the intervention effect as an adjustment to the baseline activation h_A:

$$h_A^{\text{eff}}(\boldsymbol{s}^t) = (h_A + J_{IA}) + \sum_{\text{node } j \neq I} A_{jA} J_{jA} s_j^t \tag{2.10}$$

It follows then that we can model interventions more simply as an adjustment to the baseline activation of certain beliefs and attitudes:



Let us define this more formally. For a belief system model $\mathcal{M}$ with $N \in \mathbb{N}$ nodes, and parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h} \rangle$, an *intervention*, is the function $\varphi_{\mathcal{M}} : \delta_h \mapsto \mathcal{M}'$, where $\delta_h \in \mathbb{R}^N$ is a vector of offsets to the baseline activations, and the model $\mathcal{M}'$ has parameters $\langle \boldsymbol{A}, \boldsymbol{J}, \boldsymbol{h}' \rangle$, where

$$\boldsymbol{h}' = \boldsymbol{h} + \delta_h \tag{2.11}$$

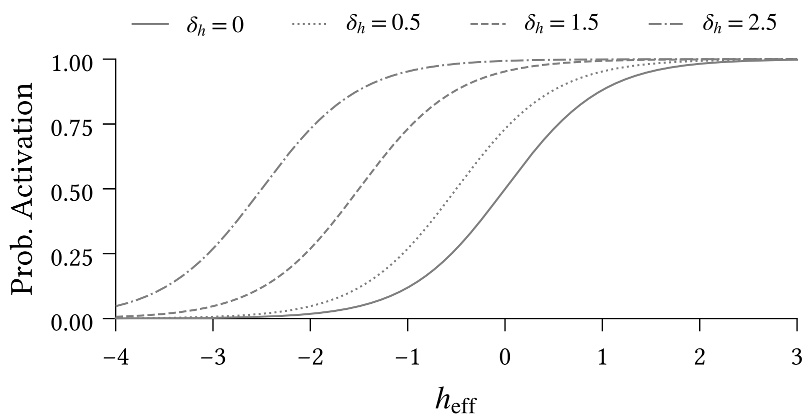


Figure 2.3: Impact of different intervention strengths ($\delta_h \in \{0, 0.5, 1.5, 2.5\}$) on activation probability for the intervention spin. Activation probability is calculated using Equation 2.6.

Figure 2.3 illustrates how the probability of activation for the intervention spin (Equation 2.6) changes for different intervention scenarios. Recall that a spin's activation probability is a function of it's effective local field. Since interventions constitute offsets to the effective local field, they are reflected as horizontal shifts in the logistic curve.

The solid line denotes the null scenario in which no intervention is applied. Consider two individuals, positioned at different locations on this base curve:

1. $h_{\text{eff}} = 0$: Ambivalent disposition toward the intervention spin, and

2. $h_{\text{eff}} = -2$: Strong negative disposition toward the intervention spin.

To understand the impacts of intervention strength on each individual, we draw a vertical line upward from the solid curve at each location, and read off the activation probability at each intersection with the other intervention curves. Under the null scenario the first individual exhibits an equal tendency toward each state, while the second is almost certain to adopt the state -1.

Notice that interventions are not experienced equivalently by the two individuals. For a weak intervention ($\delta_h = 0.5$), the first individual sees a substantial jump in activation probability, while the corresponding increase for the second individual is negligible. The stakes are reversed under a strong intervention ($\delta_h = 2.5$).

We can understand this difference in effect both conceptually and analytically. From a conceptual perspective, we can explain the large shift of the first individual under a weak intervention as resulting from their prior indifference, making them maximally susceptible to intervention. Prior to intervention, the second individual has a strong negative stance on this belief/attitude; weak interventions are insufficient to change their view. On the other hand, the second individual's negative stance provides more space for their attitude or belief to shift under a successful intervention. Compare this to the first individual, who is shifting from a place of indifference.

TODO: Quantitative explanation:

- Take derivative of the difference in activation probabilities between intervention and null scenario; solve for derivative equal to zero. Corresponds to the unique maximum since for $h_{\text{eff}} \to \pm\infty$ the second derivative is strictly positive.
- We find that maximum is at $h_{\text{eff}} = -\frac{\delta_h}{2}$.
- i.e., for a given intervention strength, the maximum change in activation probability occurs when the intervention causes h_{eff} to increase to same magnitude on the other side of zero.
- Individuals below this point require a stronger intervention to move the same amount.
- Individuals above this point experience ceiling effect. A smaller intervention could achieve almost the same effect.

Chapter 3

Methods

3.1 Counterfactual interventions

Individuals have different belief systems. Influence relations between beliefs and attitudes reflect heterogeneity in individual experiences and perception. Often these relations can themselves be considered beliefs — whether or not support for climate change mitigation leads to support for any specific policy depends in-part on an individual's belief regarding the policy's relevance. An individual's baseline activations, in turn, reflect a culmination of factors which determine their tendency toward certain beliefs or attitudes.

Our approach to parameter estimation, however, fundamentally assumes that the inferred belief system is shared by all individuals in the dataset from which the model is estimated. We note that this is not strictly a limitation of the parameter estimation method, nor of the model itself. Rather this assumption is imposed by data limitations. Fitting individual models requires substantial data from each individual, which is not available in the climate attitudes dataset. Moreover, even with substantial observational data, we are unlikely to observe all state transitions necessary to understand the *potential* dynamics of an individual system, owing to the relatively slow-moving nature of beliefs and attitudes.

For any given individual, the inferred models may thus not accurately reflect the stability of belief system states. A configuration which is stable in a given individual's own belief system may be considered unstable in the shared model. When simulating interventions, we take the most recent measurement for each individual as their initial state. However, due to this perceived instability, we expect to observe natural dynamics away from this initial state, even in null-intervention scenarios.

We thus consider intervention effects as relative to the null-intervention counterfactual baseline rather than relative to the initial state. For each intervention we run both intervention and null ($\delta_h = 0$) scenarios with identical random number generation settings, and define the *intervention effect* as the difference between the observed outcomes.

3.2 Binarisation

Prior to model fitting, we binarise all variables, mapping data values to the domain $\{-1, +1\}$. Before binarising, we shift each variable such that the *survey midpoint* (e.g., '3' on a 5-point Likert scale) aligns with 0, and rescale each variable such that the minimum and maximum possible values map to -1 and $+1$ respectively. For variables with no well-defined midpoint, such as those with an even number of possible responses, we shift such that the minimal and maximal values are at equal distance from 0.

We use a smooth binarisation process to robustly handle data points which are zero, or close to zero. This involves perturbing each data value $x \in \mathbb{R}$ by an independent noise term $\varepsilon \sim \mathcal{N}(0, \sigma)$ before thresholding. Figure 3.1 illustrates this process for a negative value of x and given choice of σ.

Observe that x is mapped to $+1$ if, and only if, ε is sufficiently large, such that $x + \varepsilon > 0$ (region A), or equivalently when $\varepsilon < x$ (region B). The probability that x is mapped to $+1$ is then

$$P(x \mapsto +1) = \Phi\left(\frac{x}{\sigma}\right) \tag{3.1}$$

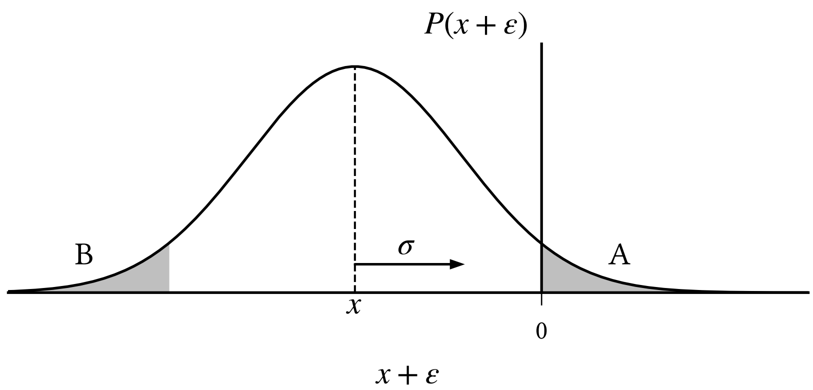


Figure 3.1: $x \in \mathbb{R}$ is smoothly binarised to $\{-1, +1\}$ by thresholding $x' = (x + \varepsilon)$ where $(x + \varepsilon) \sim \mathcal{N}(x, \sigma)$. Negative values are mapped to $+1$ with probability $P(x' > 0) = \text{A} = \text{B} = P(\varepsilon < x)$, which increases with σ and $|x|^{-1}$.

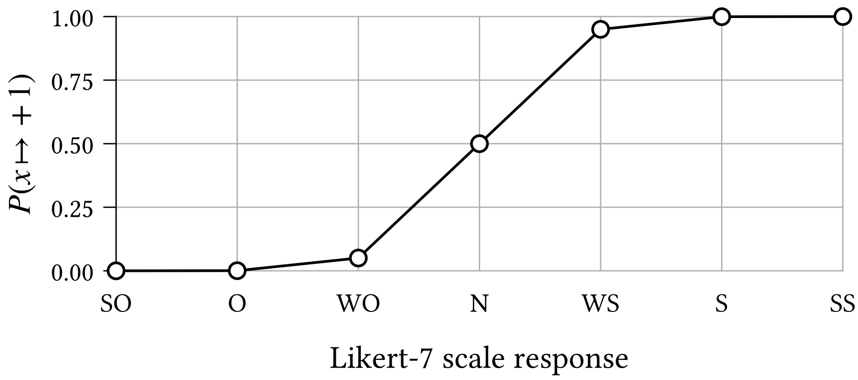


Figure 3.2: The probability of binarisation to +1 for each possible response to a Likert-7 scale survey question[2], given $\sigma \approx 0.2$.

where Φ is the standardised normal cumulative distribution function. This probability is small when x has large magnitude, or when σ is small. For the purposes of our experiments, we choose $\sigma = 0.203$ such that a 'weakly oppose' response to a 7-point Likert scale[2] (value $1/3$) is mapped to $+1$ with probability 0.05.

3.3 Parameter estimation

For the purposes of the experiments in the following sections, we perform parameter estimation to calibrate the belief system model described in Chapter 2.1 to the climate attitudes dataset (Chapter 9). We first outline the context of the parameter estimation problem, and then present our approach. We omit derivation details in this section; however, the curious reader can find these in Appendix A (**TODO**).

Consider a population of $M \in \mathbb{N}$ individuals with a shared belief system $\mathcal{M}$ comprising $N \in \mathbb{N}$ beliefs or attitudes with pre-defined adjacency matrix $\boldsymbol{A} \in \{0,1\}^{N \times N}$, for which the parameters (i.e., the baseline activations and interaction effects) are unknown. Suppose that for each individual $m \in [1, M]$ we have observed a series of measurements, reflecting that individual's belief system state at each of $t \in [1, T]$ uniformly spaced timesteps:

$$\left\{\boldsymbol{x}^t_{(m)}\right\}_{t=1}^{T}, \quad \text{where each } \boldsymbol{x}^t_{(m)} \in \mathbb{R}^N \tag{3.2}$$

The measurements need not be binary, but are assumed to be real. The collection of measurements across all individuals forms a dataset D, which is the particular observed value of the random variable $\mathcal{D}$ representing possible datasets.

[2]The oppose/support 7-point Likert scale has possible responses: strongly oppose, oppose, weakly oppose, neutral, weakly support, support, strongly support.

While maximum likelihood estimation (MLE) is often used to infer parameters for similar models (Lee et al., 2015; Nguyen et al., 2017), we should be cautious, for two reasons, of applying this method naïvely in the present study.

Firstly, the model defined in Equation 2.7 assumes binary spin states $s \in \{-1, +1\}$. The climate attitudes dataset does not satisfy this assumption and must be binarised for MLE to be applicable. However, we have no guarantee that the parameterisation inferred for any specific binarisation is a reasonable explanation for the non-binarised dataset, nor for any other possible binarisation in the case where a probabilistic binarisation scheme is used (such as the one described in Chapter 3.2).

Secondly, maximum likelihood estimation is prone to overfitting when the number of model parameters is similar to the number of observations (Epskamp, Borsboom, et al., 2018). While we consider only a small number of beliefs and attitudes in this study ($N \in \{7, 8\}$), the number of parameters p grows rapidly, with $p \in O(N^2)$ for both the symmetric and asymmetric model variants. In combination with sampling error, we expect the number of parameters to negatively impact the robustness of the inferred parameters — given an alternative dataset of the same size, we would expect significant variation in parameters.

We mitigate these respective issues in our parameter estimation scheme by (i) marginalising over the set of possible binarisations of D, and (ii) applying regularisation based on parameter magnitudes. We choose the parameterisation $\boldsymbol{\theta}^* \in \mathbb{R}^p$ which satisfies the optimisation problem:

$$\boldsymbol{\theta}^* = \operatorname*{argmax}_{\hat{\boldsymbol{\theta}} \in \mathbb{R}^p} f\big(D; \hat{\boldsymbol{\theta}}\big), \quad \text{where} \quad f\big(D; \hat{\boldsymbol{\theta}}\big) \coloneqq \mathcal{L}_D\big(\hat{\boldsymbol{\theta}}\big) - \lambda \sum_{\theta \in \hat{\boldsymbol{\theta}}} \sqrt{\theta^2 + \varepsilon} \tag{3.3}$$

The first term in the objective function f, $\mathcal{L}_D\big(\hat{\boldsymbol{\theta}}\big)$, denotes the *expected* log-likelihood given a parameterisation $\hat{\boldsymbol{\theta}}$, over possible binarisations of D:

$$\mathcal{L}_D\big(\hat{\boldsymbol{\theta}}\big) = \mathbb{E}\Big[L_{D_B}\big(\hat{\boldsymbol{\theta}}\big) \mid \mathcal{D} = D\Big] = \sum_{D_B} P(\mathrm{Bin}(\mathcal{D}) = D_B \mid \mathcal{D} = D) \cdot L_{D_B}\big(\hat{\boldsymbol{\theta}}\big) \tag{3.4}$$

where $L_{D_B}\big(\hat{\boldsymbol{\theta}}\big)$ is the log-likelihood of a specific binarisation D_B. We will defer explicitly defining $L_{D_B}\big(\hat{\boldsymbol{\theta}}\big)$ for now, but will return to this point shortly in Equation 3.5.

TODO: Intuition for the effect of using the expectation.

- Inferred model reflects ambiguity in states close to zero.
- The likelihood contribution of each data value is the expectation over possible spin states.

- Values close to zero contribute a comparable weighting from each binary state.
- When the likelihood given one binary state would be significantly higher than for the other, this is downweighted to account for the probability that the value does not obtain this state.
- Therefore the expectation prevents overconfident optimisation toward any specific binarisation — the inferred model is required to explain the binarised dataset *in expectation*.
- In other words, if a data value may be binarised to -1 or $+1$ with equal probability, then both cases should be reasonably explained by the parameterisation.

The second term is a smooth variant of L1 regularisation (Tibshirani, 1996). This has the effect of penalising nonzero parameters which do not meaningfully contribute to the expected likelihood. Parameters instead must reflect relationships which are sufficiently prevalent in the data, reducing the robustness issues resulting from sampling error and a large number of parameters. The hyperparameter $\lambda \in \mathbb{R}^+$ controls regularisation strength, with higher values resulting in sparser models. As $\varepsilon \to 0^+$ the second term converges to the standard formulation of L1 regularisation. Unlike the standard formulation, when $\varepsilon > 0$ the first-derivative of this term exists, so is amenible to gradient-based optimisation. We discuss the choice of values for λ and ε in Chapter 3.3.1.

Let $D_B \sim \mathrm{Bin}(D)$ be a possible binarisation of D, and $\boldsymbol{\theta}$ a parameterisation for the (asymmetric or symmetric) belief system model, which is decomposable into the model parameters $\boldsymbol{\theta} = \langle \boldsymbol{J}, \boldsymbol{h} \rangle$. The log-likelihood of D_B given $\boldsymbol{\theta}$ is:

$$\begin{aligned} L_{D_B}(\boldsymbol{\theta}) &= \log P(D_B \mid \boldsymbol{\theta}) \\ &= \sum_{m=1}^{M} \sum_{t=1}^{T-1} \log s_{i,(m)}^{t+1} \cdot h_i^{\mathrm{eff}}\Big(\boldsymbol{s}_{(m)}^{t}\Big) - \log\Big(2 \cosh h_i^{\mathrm{eff}}\Big(\boldsymbol{s}_{(m)}^{t}\Big)\Big) \end{aligned} \tag{3.5}$$

The derivation of Equation 3.5 is analogous to that of the non-equilibrium Ising model log-likelihood (Nguyen et al., 2017). Combining Equation 3.5 and Equation 3.4, we obtain an explicit expression for the expected likelihood:

$$\mathcal{L}_D(\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \left(\sum_{m=1}^{M} \sum_{t=1}^{T-1} P\Big(\boldsymbol{S}_{(m)}^{t} = \boldsymbol{s} \mid D\Big) \cdot \mathbb{E}\Big[\boldsymbol{S}_{(m)}^{t+1} \mid D\Big]^T h^{\mathrm{eff}}(\boldsymbol{s}) \right) - Z(D; \boldsymbol{\theta}) \tag{3.6}$$

where $\mathbb{E}\Big[\boldsymbol{S}_{(m)}^{t+1} \mid D\Big]$ is the expected binary configuration for the observation $\boldsymbol{x}_{(m)}^{t+1}$, the vector $h^{\mathrm{eff}}(\boldsymbol{s})$ contains the effective baseline activation for each spin given the previous state $\boldsymbol{s}$, and Z is the expected log partition function, summed across observations:

$$Z(D;\boldsymbol{\theta}) = \sum_{\boldsymbol{s}} \sum_{\substack{m \le M \\ t < T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \sum_{i=1}^{N} \log\big(2\cosh h_i^{\text{eff}}(\boldsymbol{s})\big) \tag{3.7}$$

Per the optimisation problem defined in Equation 3.3, we choose the parameterisation $\boldsymbol{\theta}^*$ which maximises the regularised expected log-likelihood, which occurs when $\frac{\partial f}{\partial \theta} = 0$ for every parameter $\theta \in \boldsymbol{\theta}^*$. The partial derivative with respect to a given baseline activation parameter h_i, for $i \in [1, N]$, is given by:

$$\begin{aligned} \frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta}) = &\sum_{\boldsymbol{s}} \sum_{\substack{m \le M \\ t < T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \Big[\mathbb{E}\Big[S^{t+1}_{i,(m)} \mid D\Big] - \tanh\big(h_i^{\text{eff}}(\boldsymbol{s})\big)\Big] \\ &- \frac{\lambda h_i}{\sqrt{h_i^2 + \varepsilon}} \end{aligned} \tag{3.8}$$

Note that the expressions for $\mathcal{L}_D(\boldsymbol{\theta})$ and $\frac{\partial}{\partial h_i} f(D;\boldsymbol{\theta})$ are applicable to both the asymmetric and symmetric belief system models. This property does not hold for the partial derivatives with respect to interaction effects. In the asymmetric model, for $i, j \in [1, N]$, this is derived analogously to Equation 3.8 as:

$$\begin{aligned} \frac{\partial}{\partial J_{ji}} f(D;\boldsymbol{\theta}) = &\sum_{\boldsymbol{s}} \sum_{\substack{m \le M \\ t < T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) A_{ji} s_j \Big[\mathbb{E}\Big[S^{t+1}_{i,(m)} \mid D\Big] - \tanh\big(h_i^{\text{eff}}(\boldsymbol{s})\big)\Big] \\ &- \frac{\lambda J_{ji}}{\sqrt{J_{ji}^2 + \varepsilon}} \end{aligned} \tag{3.9}$$

In the symmetric belief system model, since we require that $\boldsymbol{J} = \boldsymbol{J}^T$, we estimate a single interaction parameter $J_{ji} = J_{ij} = \beta_{\{i,j\}}$ for each pair of spins S_i, S_j. Each pair thus contributes twice to the partial derivative, once for each direction:

$$\begin{aligned} \frac{\partial}{\partial \beta_{\{i,j\}}} f(D;\boldsymbol{\theta}) = &\sum_{\boldsymbol{s}} \sum_{\substack{m \le M \\ t < T}} P\Big(\boldsymbol{S}^t_{(m)} = \boldsymbol{s} \mid D\Big) \sum_{(i',j') \in \{i,j\}} A_{j'i'} s_{j'} \Big[\mathbb{E}\Big[S^{t+1}_{i',(m)} \mid D\Big] - \tanh\big(h_{i'}^{\text{eff}}(\boldsymbol{s})\big)\Big] \\ &- \frac{\lambda \beta_{\{i,j\}}}{\sqrt{\beta_{\{i,j\}}^2 + \varepsilon}} \end{aligned} \tag{3.10}$$

3.3.1 Hyperparameters

The optimisation problem in Equation 3.3 has two hyperparameters which must be specified prior to parameter estimation: the regularisation strength $\lambda \in \mathbb{R}^+$, and the smoothing parameter $\varepsilon \in \mathbb{R}^+$. The values for both are summarised in Table 3.1.

Parameter	Model type	Dataset	Value
λ	Symmetric	Full	1.4×10^{-2}
	Asymmetric	Full	1.2×10^{-2}
		Conservative	3.4×10^{-2}
		Liberal	4.3×10^{-2}
ε	All	All	10^{-8}

Table 3.1: Values for regularisation strength (λ) and smoothing (ε) hyperparameters. Regularisation strength model- and dataset-specific; *Conservative* and *Liberal* refer to subsets of the climate attitudes dataset comprising individuals with the specified ideology (Chapter 9).

We use the Extended Bayesian Information Criterion (EBIC) (Chen & Chen, 2008) to select λ, as recommended by Epskamp, Borsboom, et al. (2018):

$$\mathrm{EBIC}(\boldsymbol{\theta}^*) = \overbrace{k \ln q - 2\mathcal{L}_D(\boldsymbol{\theta}^*)}^{\mathrm{BIC}(\boldsymbol{\theta}^*)} + 2k \cdot \gamma \ln p \tag{3.11}$$

The variable k denotes the number of nonzero parameters in the optimised model, where a parameter θ is considered nonzero if $|\theta| > 10^{-2}$. The point of difference between the EBIC and the standard Bayesian Information Criterion (BIC) is in their assumed priors over the space of possible models. While the BIC assumes a uniform prior, the EBIC penalises values of k which permit a large number of models (Barber & Drton, 2015; Foygel & Drton, 2010). For instance, the class of models comprising eight spins (i.e., with 72 parameters) contains considerably fewer models with 20 parameters (5.4×10^{11}) than models with 36 parameters (4.4×10^{20}). In this scenario, by assuming a uniform prior, the BIC implicitly penalises models which are too sparse, or too dense. The paramater $\gamma \geq 0$ specifies the degree to which the EBIC penalises such classes of models, with $\gamma = 1$ corresponding to a uniform prior over models with k parameters[3], for each $k \in 0..p$.

Barber & Drton (2015) demonstrate that small, nonzero choices of γ are sufficient to improve the accuracy of Ising model structural inference in various contexts. We take $\gamma = 0.25$, which is the minimum value considered in their study. We select λ which minimises the EBIC for the symmetric and asymmetric models (independently), the results of which are displayed in Figure 3.3.

[3]Note that this is different from the prior used in the BIC, which is uniform over the entire space of models. The prior in the EBIC, for $\gamma = 1$, is uniform only within each class of models containing k parameters.

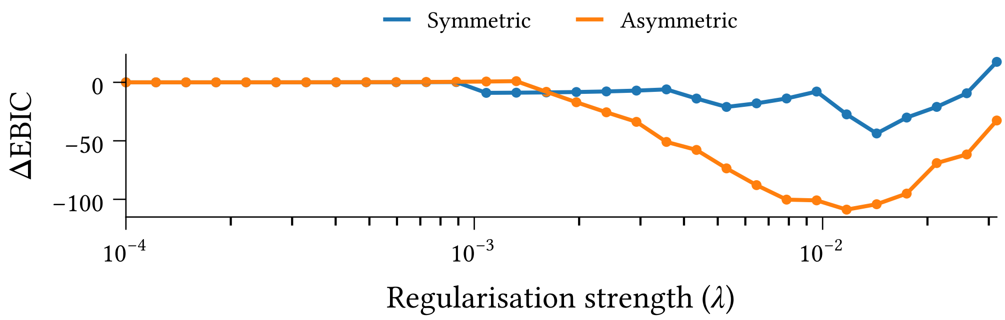


Figure 3.3: Effect of regularisation strength λ on Extended Bayesian Information Criterion for symmetric and asymmetric belief system models optimised to the climate attitudes dataset using Equation 3.3. The vertical axis measures the difference in EBIC compared to the no-regularisation case ($\lambda = 0$).

We consider parameters with magnitude less than 10^{-2} as 'effectively zero', and take $\varepsilon = 10^{-8}$ such that $\sqrt{\varepsilon}$ is significantly smaller than this threshold. This choice of ε ensures that the smooth regularisation remains a good approximation to L1 regularisation, i.e.,

$$\sqrt{\theta^2 + \varepsilon} \approx |\theta| \tag{3.12}$$

3.3.2 Implementation details

We solve the parameter estimation problem using the Scipy 1.17.1 implementation of the quasi-Newton BFGS optimisation algorithm (Nocedal & Wright, 2006, p. 136; Virtanen et al., 2020). We use the analytic jacobian comprising the partial derivatives stated above. We take $\boldsymbol{\theta} = \mathbf{0}$ as the initial guess.

To ensure numerical stability irrespective of dataset size, we rescale the expected log-likelihood contribution to the objective function and partial derivatives, dividing by the number of observed observations (time intervals) in the dataset. For $M \in \mathbb{N}$ individuals and $T \in \mathbb{N}$ timesteps this amounts to a scale factor of $\frac{1}{M(T-1)}$.

Chapter 4

Model Calibration

TODO:

- Discuss differences in baseline activations.
  - Exogenous influences, and unmeasured beliefs

The **climate beliefs dataset**, detailed in Chapter 9, comprises eight beliefs and attitudes relating to climate change (Table 4.1), extracted from the CCCV survey (cf. Constantino et al. (2022). The dataset includes responses from 1693 repeating participants, measured during waves 3 and 4 of the survey. We map the maximum and minimum (possible) values for each variable to $+1$ and -1 respectively, such that these reflect the two spin states in the KBS model.

Before proceeding, let's take a moment to calibrate and evaluate the models which will be used in the upcoming experiments. Using the parameter estimation method outlined in the previous chapter, we calibrate the symmetric and asymmetric belief system models to the climate beliefs dataset (see Chapter 9). We will evaluate the calibrated models on both structural accuracy ('how accurate are the parameter estimates?') and predictive capacity ('how well do the models explain the data?').

Figure 4.2 shows the interaction effect matrix, $\boldsymbol{J}$, for each model. We only display the upper triangular portion of the symmetric model's matrix, since each pair of spins in this model has a single, bidirectional relation. Both models feature a dominant diagonal, indicating that most variables are slow-moving with respect to the modelled timescale[4] (they are 'sticky'), which is consistent with prior studies looking at the rate of change for beliefs/attitudes (Green & Platzman, 2024; Kiley & Vaisey, 2020). This is particularly true for `Politics`, and less so for `CC Others Worry` and `CC Impact`.

[4]The models' timescales are set by the duration between survey responses, in this case approximately six months (Chapter 9).

Item	Index	Interpretation
CC Real	No	Belief that climate change is/is not real.
CC Human	No	Belief that climate is/is not caused (at least in-part) by human activities.
CC Worry	No	Level of worry about current and future climate change.
CC Others Worry	No	Belief regarding *others'* level of worry about current and future climate change.
Weather Worry	No	Level of worry about possible near-term extreme weather events or natural disasters.
Politics	Yes	Political views, as a combination of partisan alignment and political ideology.
CC Impact	Yes	Belief regarding the degree of current climate change impacts, in general.
CC Action	Yes	General attitude toward action on climate change.

Table 4.1: Variables included in the climate beliefs dataset. Index variables are constructed by taking the average of their constituent columns, after re-scaling to the interval $[-1, 1]$. Note that this table is identical to Table 9.2 displayed in Chapter 9.

Observe that all interaction effects are non-negative. This is by design; we have re-coded the dataset variables such that this is the case. The fact that such a re-coding exists implies that it is possible, within this belief system, to hold a set of beliefs and attitudes which are internally consistent, i.e., with no cognitive dissonance. We provide a formal proof of this statement in Appendix A.

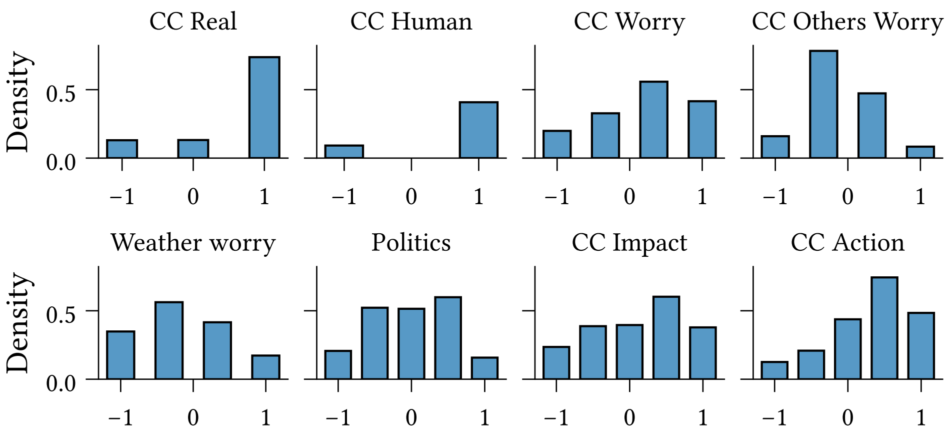


Figure 4.1: Marginal distribution for each of the eight variables in the climate beliefs dataset (Table 4.1). Note that this figure is identical to Figure 9.7 displayed in Chapter 9.

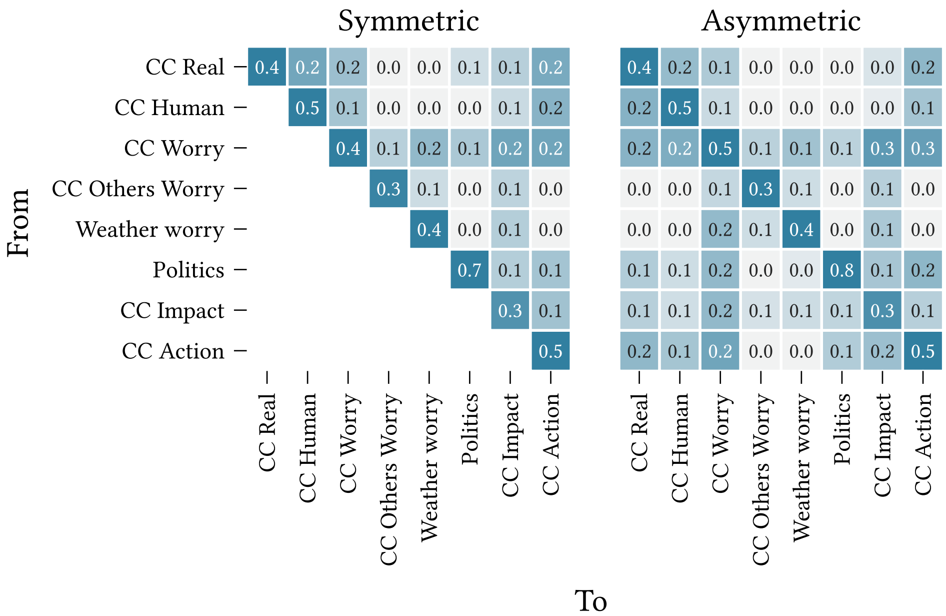


Figure 4.2: Interaction effect matrices ($\boldsymbol{J}$) for the symmetric and asymmetric belief system model variants, calibrated to the climate beliefs dataset (Chapter 9) using the parameter estimation method in Chapter 3.3.

As recommended in Epskamp, Borsboom, et al. (2018), we evaluate interaction parameter accuracy by examining the bootstrapped confidence intervals around each parameter estimate. We use bootstrapping to estimate the uncertainty in our parameter estimates due to sampling error. Let $\boldsymbol{D} \in \mathbb{R}^{M \times T \times N}$ be the complete dataset, where M, T, and N denote the number of participants, observations, and spins respectively[5]. We construct

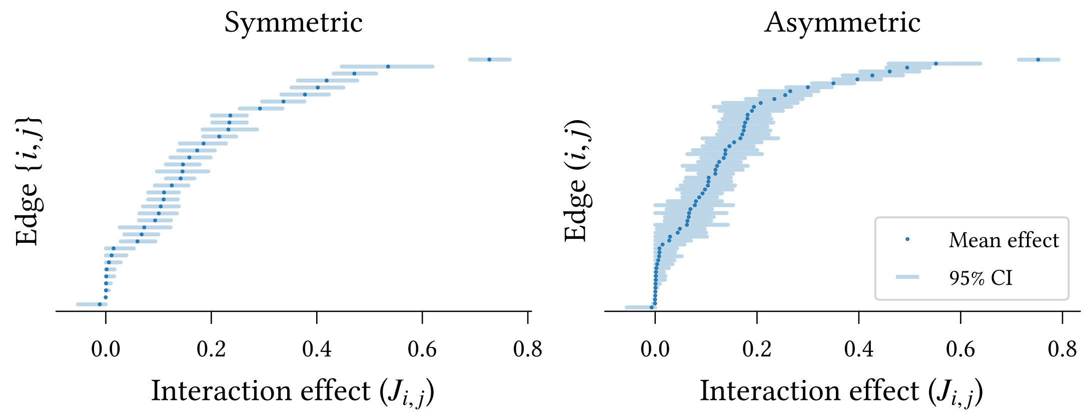


Figure 4.3: Interaction effect parameter accuracy for symmetric and asymmetric models calibrated to the climate beliefs dataset (Chapter 9). Mean parameter values are shown in increasing order. 95% confidence intervals are calculated using the percentile method over models calibrated to bootstrapped datasets (500 repeats).

[5]In the climate beliefs dataset we have $M = 1693$, $T = 2$, and $N = 8$

500 bootstrapped datasets by sampling rows (participants) with replacement from $\boldsymbol{D}$, such that each bootstrapped dataset has the same shape as the complete dataset.

For each bootstrapped dataset $\boldsymbol{D}_{(i)}$ we calibrate a model $\mathcal{M}_{(i)}$ with parameters

$$\langle \boldsymbol{A}, \boldsymbol{J}_{(i)}, \boldsymbol{h}_{(i)} \rangle =: \boldsymbol{\theta}^{*}_{(i)} \in \mathbb{R}^{p} \tag{4.1}$$

where $p \in \mathbb{N}$ is the number of model parameters, and we take the adjacency matrix $\boldsymbol{A}$ as fully-connected, permitting influence relations to be inferred between each pair of beliefs or attitudes.

Figure 4.3 shows the estimated interaction effects in increasing order for each model. The confidence intervals are calculated using the percentile method (**CITE**) across the bootstrapped models' parameters. Most parameters display small 95% confidence intervals, indicating an accurate fit for both models. On average, the symmetric model exhibits smaller confidence intervals than the asymmetric model ($0.065 < 0.085$), which is expected given the larger number of parameters in the asymmetric model. In both models the self-interaction effect on `Politics` is significantly larger than all other parameters, as evidenced by the non-overlapping confidence intervals, supporting our earlier observation regarding Figure 4.2. Moreover, the remaining self-interaction effects are, in-general, significantly larger than the pairwise effects, with the exception of `CC Impact` in the symmetric model, and `CC Impact`, `CC Worry Others` in the asymmetric model.

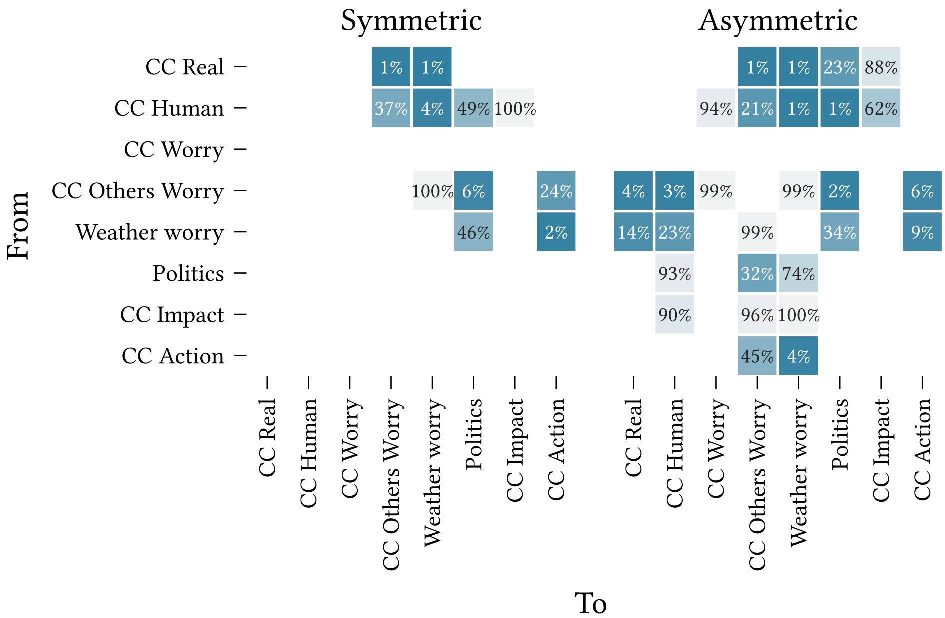


Figure 4.4: Edge selection probability for models calibrated to bootstrapped datasets sampled with replacement from the climate beliefs dataset (500 repeats). Edges are 'selected' if they survive regularisation with magnitude at least 0.01.

Due to the use of regularisation in model calibration we must be careful not to draw conclusions regarding edge *existence* from this figure (Epskamp, Borsboom, et al., 2018). Instead, we may consider the proportion of bootstrapped models for which each edge is nonzero, shown in Figure 4.4. Relations which are selected in all models are not shown.

All edges excluded from the model calibrated on the full dataset (Figure 4.2) have low selection probability in the bootstrapped datasets ($< 50\%$). The relation `CC Real` $\rightarrow$ `Politics` is excluded from the complete asymmetric model but included (bidirectionally) in the symmetric model. This is reflected in the corresponding selection probabilities—this edge is selected in 100% of bootstrapped symmetric models but only 23% of asymmetric models. Edges which *are* selected in the complete models generally have high selection probability. The notable exception in the asymmetric case, `CC Human` $\rightarrow$ `CC Impact`, has a relatively small effect size ($J_{i,j} \approx 0.04$).

We now evaluate the models' predictive capacities[6]. First, we examine the reliability of predicted transitions. Figure 4.5 shows the average binarisation probability for each spin (i.e., the probability that the corresponding observation in the second wave of the dataset is binarised to +1), binned by transition probability. Bins with no observations are not shown (e.g., extreme values for `CC Others Worry`).

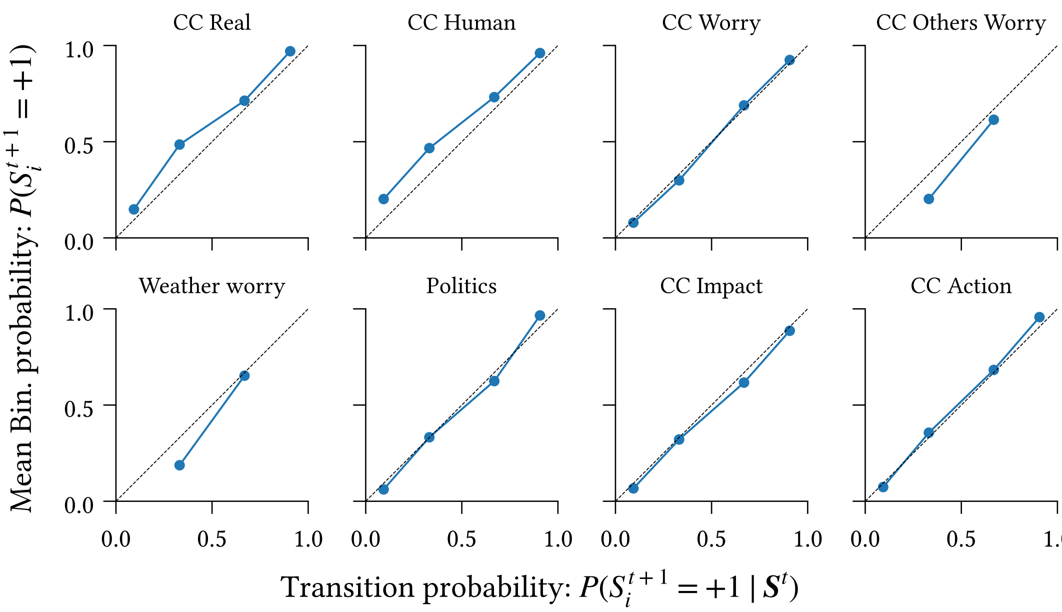


Figure 4.5: Reliability of transitions predicted by the model measured as the agreement between conditional probabilities under the model, $P\big(S_i^{t+1} = +1 \mid \boldsymbol{S}^t\big)$, and expected proportion of +1 values after binarisation, calculated as the average transition probability for survey participants in a given conditional probability bin.

[6]While we only show results for the asymmetric model here, the corresponding figures for the symmetric model are substantially similar, and all conclusions drawn regarding the asymmetric model apply also to the symmetric one.

The two measurements are strongly correlated for most variables. The higher variation for `CC Real` and `CC Human` likely reflects the observed heavy skew in these variables toward larger values (see Figure 9.7 in Chapter 9). We observe no high/low probabilities for either `CC Others Worry` or `Weather Worry`, on account of the limited influence of other spins on these variables, as seen in the corresponding columns of Figure 4.2.

Next, we compare the calibrated model against a null model, in which we permit only baseline activation (h_i) and self-influence ($J_{i,i}$) parameters. We measure the relative entropy of the binarised data from the second wave of the dataset, with respect to the probability distributions induced by each model. For each participant and spin, we calculate the relative entropy as

$$D(P \parallel Q) \coloneqq H_C(P,Q) - H(P) \tag{4.2}$$

Where $P \coloneqq P\Big(\mathrm{Bin}\Big(X^{t+1}_{i,(m)}\Big)\Big)$ is the distribution over binarisations of the observation in the first timestep, and $Q \coloneqq P_{\mathcal{M}}\Big(S^{t+1}_{i,(m)}\Big)$ is the distribution over subsequent states according to the model, marginalising over binarisation of the previous observation,

$$Q \coloneqq P_{\mathcal{M}}\Big(S^{t+1}_{i,(m)}\Big) = P_{\mathcal{M}}\Big(S^{t+1}_{i,(m)} \mid \mathrm{Bin}\Big(X^{t}_{i,(m)}\Big)\Big) \cdot P\Big(\mathrm{Bin}\Big(X^{t}_{i,(m)}\Big)\Big) \tag{4.3}$$

The functions H and H_C denote the entropy and cross-entropy, respectively. The entropy term quantifies the uncertainty in the binarisation process, and the cross-entropy measures the 'expected surprise' when comparing predictions generated from the model with the true binarised dataset. Formally, these are defined as follows:

$$H(P) = -\sum_{s\in\pm1} P(s)\log_2 P(s) \tag{4.4}$$

$$H_C(P,Q) = -\sum_{s\in\pm1} P(s)\log_2 Q(s) \tag{4.5}$$

The relative entropy, $D(P \parallel Q)$, is small when either (i) the model accurately predicts the observed binarised value such that the cross-entropy is low[7], or (ii) the binarisation process is very uncertain. It follows that the relative entropy is high in cases where the model fails to accurately predict the next state, despite the binarisation process being fairly deterministic.

[7]The relative entropy is strictly non-negative (**CITE**), so the binarisation entropy is a lower bound on the cross-entropy. This aligns with our intuition—the model cannot possibly be highly accurate when the binarisation process is very uncertain.

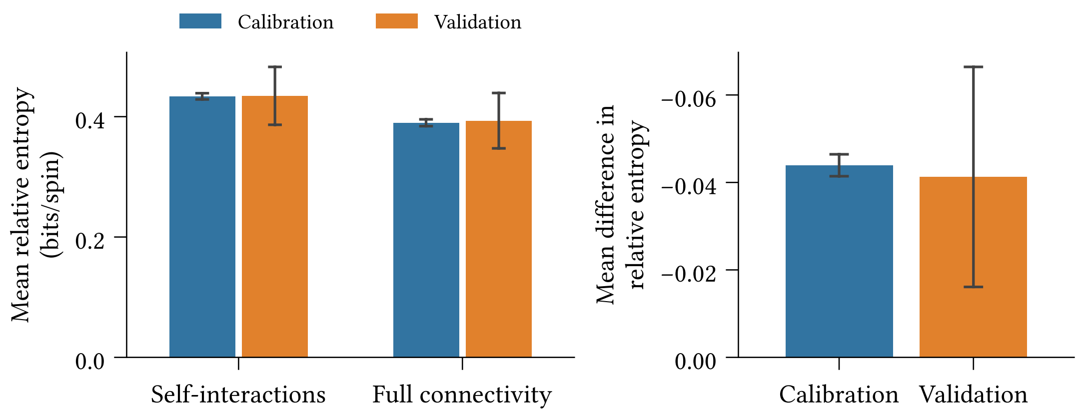


Figure 4.6: *(Left)* Mean absolute relative entropy of the binarised dataset with respect to the calibrated model, for the full-connected and null (only self-interactions) models, using 10-fold cross-validation. Average calculated across spins and survey participants. *(Right)* Mean difference in relative entropy. Confidence intervals display two standard deviations around the mean value.

To test for the effect of including cross-interactions, we fit the fully-connected model and null models using 10-fold cross-validation. For each survey participant in the validation (holdout) split we calculate the mean difference in relative entropy between the fully-connected and null models. Figure 4.6 shows the mean relative entropy across individuals and spins on the null and fully-connected models (left) and the mean *difference* in relative entropy, averaged across survey participants (right).

We find, in the right-hand panel, that the fully-connected model leads to significant reduction in relative entropy averaged over survey participants ($p < 0.05$). While statistically significant, the reduction is small. Since the measurement is averaged across survey participants, this finding suggests that the self-interaction model is sufficient to explain the behaviour for most participants—reflecting the slow-moving dynamics of the observed belief system—but not all participants. We anticipate that individuals whose belief states change more between observations are better explained by the fully-connected model than the null model.

Figure 4.7 explores this hypothesis, displaying the empirical probability density and cumulative distribution functions for the mean difference in relative entropy, across individuals. Indeed, we see substantial variation in effects. The weak right tail and heavy left tail indicate cases which are better-explained by the null and fully-connected models respectively.

We examine a sample of cases from either tail (top panel: left tail, bottom panel: right tail) in Figure 4.8. Each panel displays the change in binarisation probability between the two observed survey waves for a sampled survey participant. In the first three panels

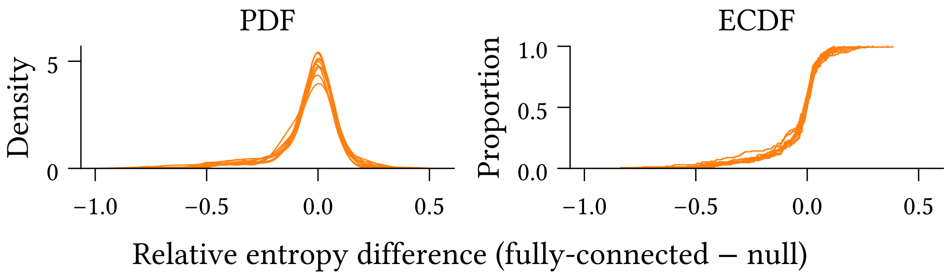


Figure 4.7: Probability density function (left) and empirical cumulative distribution function (right) of the mean difference in relative entropy between the fully-connected and null (only self-interaction) models across survey participants. Negative values indicate lower relative entropy for the fully-connected model.

of the top row, we observe scenarios in which most spins are initially aligned, and a subset of the remaining spins then update to align with this set, i.e., where the system shifts toward a more consistent state. Conversely, in the first, second, and fourth panels in the bottom row we see the opposite scenario play out. That is, the system is initially well-aligned, yet some spins update to a less consistent state.

The observed behaviour aligns with our expectations, namely that the model with cross-interactions outperforms the null model in scenarios where the belief system state updates toward a more consistent state (according to the theory of cognitive dissonance), and is out-performed when participants' behaviour contradicts this theory.

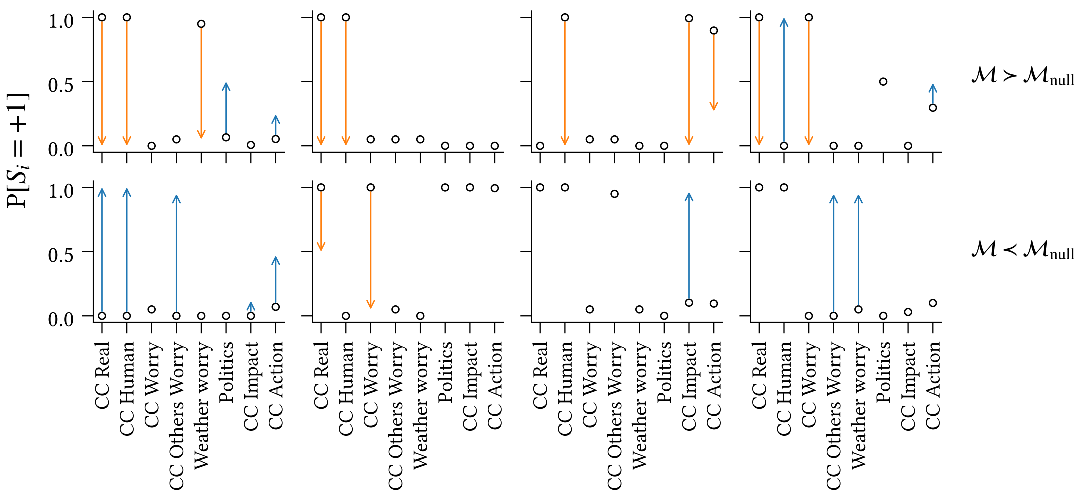


Figure 4.8: Observed transitions for sample survey participants in the left tail (top row) and right tail (bottom row) of Figure 4.7. Circles indicate the probability that participants' observations in the first wave are binarised to +1. Arrows denote the change in the second wave, colour-coded according to whether the change is toward the positive (blue) or negative (orange) state.

Chapter 5

Existence and impact of asymmetry in belief systems

Belief system models based on the Ising model, inspired by the cognitive dissonance theory of belief system dynamics, demonstrate impressive descriptive and explanatory capacities for various observed cognitive phenomena at both individual and collective scales. The currently prevalent view within the modelling community is that influence relations between beliefs or attitudes are symmetric — the pressure for X to align with Y is equal to that placed on Y to align with X. The present chapter places this assumption on trial.

We begin, in Chapter 5.1, by calibrating the non-equilibrium belief system model to the climate beliefs dataset described in Chapter 9.3. We demonstrate that belief system relations cannot be universally classified as symmetric, that some influence relations are recognisably asymmetric, and that relational asymmetry can vary in degree. In Chapter 5.2, we then show that this finding is not inconsequential. Asymmetric models exhibit different dynamics in experiments on belief-level interventions, with the potential to affect conclusions drawn about the relative effectiveness of interventions.

5.1 Existence of asymmetric relations

TODO:

- Consider adding squares to the asymmetric entries in Figure 5.2

To investigate the existence of asymmetric relations in the asymmetric model calibrated to the climate beliefs dataset, we examine the differences in directional interaction effects for the asymmetric models calibrated using bootstrapping in the previous section.

For each bootstrapped model, $\mathcal{M}_{(i)}$ with parameters $\langle \boldsymbol{A}, \boldsymbol{J}_{(i)}, \boldsymbol{h}_{(i)} \rangle$, we obtain an estimate for the directional differential matrix:

$$\Delta_J^{(i)} = \boldsymbol{J}_{(i)} - \boldsymbol{J}_{(i)}^T \tag{5.1}$$

Recall that the k'th row of $\boldsymbol{J}_{(i)}$ (for $k \in \{1, ..., N\}$) describes the strength and direction of influence *from* the spin S_k *toward* each other spin. Hence for $k, \ell \in \{1, ..., N\}$ we should interpret the element $\Delta_J^{(i)}|_{k,\ell}$ of the directional differential matrix as the excess influence of spin S_k on S_ℓ. A positive value indicates that S_k exerts greater influence on S_ℓ than S_ℓ does on S_k.

Figure 5.1 shows the median directional differential for each pair of spins in the asymmetric model, in decreasing order. The 90% confidence intervals are calculated as the 5th and 95th percentiles across bootstrap samples of the directional differential

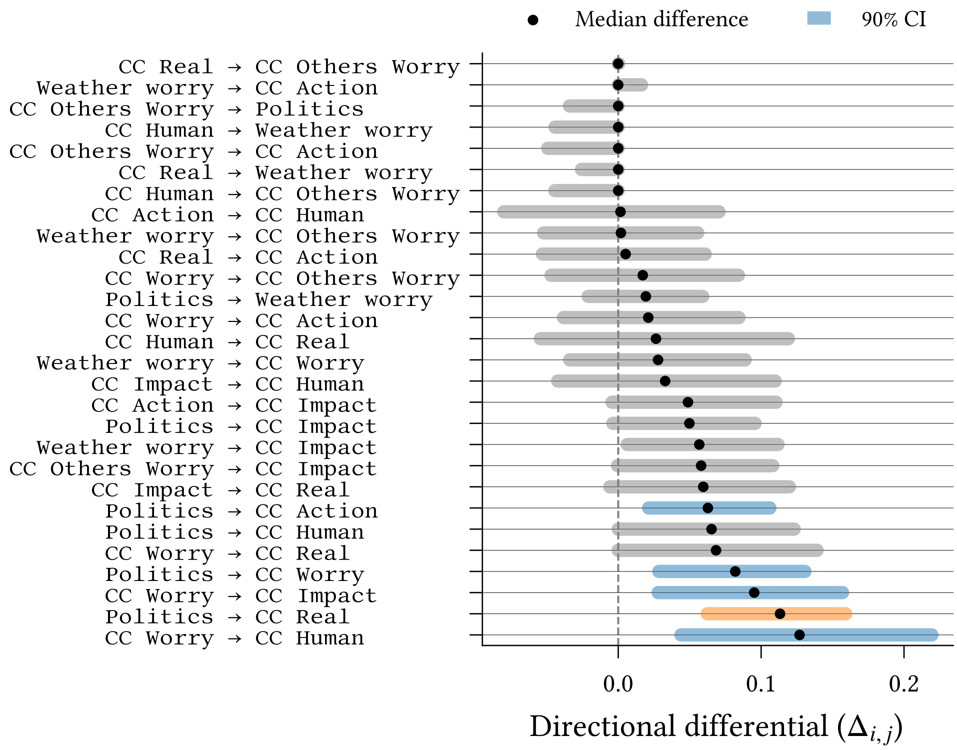


Figure 5.1: Directional differentials—the difference between directed interaction effects for a pair of spins—in reverse order of median value. Confidence intervals are calculated using the percentile method on models calibrated using bootstrapping ($n = 500$). Grey confidence intervals indicate absence of significant asymmetry. For cases with significant asymmetry, the colour indicates that one (**orange**) or both (**blue**) of the directed interacton effects are nonzero.

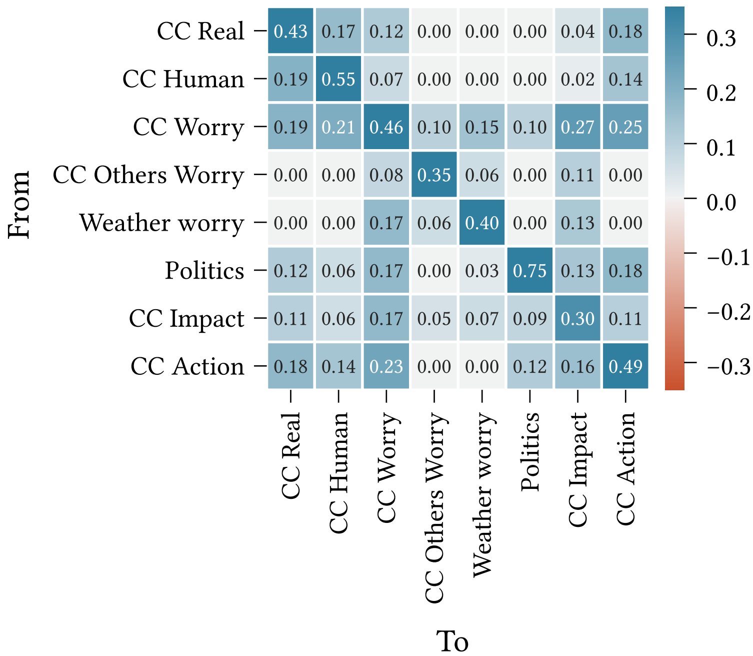


Figure 5.2: Interaction effect matrix ($\boldsymbol{J}$) for the asymmetric belief system model calibrated to the full climate beliefs dataset. This figure is identical to the asymmetric interaction matrix in the previous chapter (Figure 4.2).

matrix elements[8]. For each pair of spins we display only the directional differential for which the median is positive (since Δ_J is symmetric), and we exclude diagonal entries (which are zero by definition).

Figure 5.1 shows a majority of symmetric relations between spins (characterised by confidence intervals containing zero), with a small number of asymmetric relations. We can partition the asymmetric relations into two groups, according to whether directional interactions exist in only one direction, indicated using orange confidence intervals, or both directions with different strengths, indicated with blue confidence intervals (see Figure 5.2, identical to the interaction matrix shown in Figure 4.2 of the previous chapter). Pairs with two nonzero interactions comprise `Politics` $\rightarrow$ {`CC Action`, `CC Worry`} and `CC Worry` $\rightarrow$ {`CC Impact`, `CC Human`}; the only pair with a single directional interaction is `Politics` $\rightarrow$ `CC Real`.

Comparing directional strength and degree centrality indices for `CC Worry` and `Politics` across bootstrapped models, we find significant differences ($p < 0.05$) for strength centrality on `CC Worry`, and both strength and degree centrality on `Politics`.

[8]In Chapter 4 we cautioned against using bootstrapped confidence intervals to test for the *existence* of edges by comparison with zero when regularisation is used during calibration—this caution does not apply to the *comparison* of edge weights via the mean difference (Epskamp, Borsboom, et al., 2018).

Most directional differentials display substantial uncertainty (confidence interval width > 0.1 on average), reflecting the parameter accuracy discussed in the previous chapter (Figure 4.3). This results in several inconclusive cases, where the median bootstrap-estimate directional differential is nonzero but the confidence interval contains zero. In other cases we observe apparent symmetric or near-symmetric relations (e.g., `Weather worry` $\longrightarrow$ `CC Others Worry`).

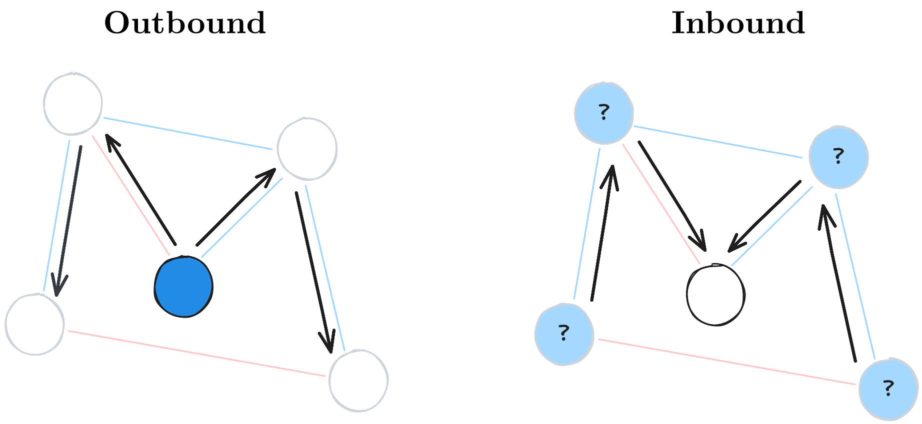
At the top of Figure 5.1 we see several apparently skewed confidence intervals. These all correspond to cases where both interactions are zero in the complete model (Figure 5.2), so the skews likely reflect differences in edge selection probability between the two directional interactions (Figure 4.4). For instance, `CC Others Worry` $\longrightarrow$ `CC Human` exhibits a larger apparent skew than `CC Others Worry` $\longrightarrow$ `CC Real`, and this is reflected in observed differences in selection probability.

Due to the large confidence intervals on most directional differentials we are less likely to observe significant asymmetric relations between variables with small interaction effects (since the possible difference in effects is smaller by definition). Indeed, all observed significant cases of asymmetry have at least one interaction effect with magnitude exceeding 0.1. This raises the question of whether asymmetry is instead simply explained by a high total interaction influence in combination with sampling error. However, this is disputed by `CC Action`, which has several strong interaction effects (its outbound effects are typically larger than those of `Politics`), yet does not exhibit significant asymmetric influence over any other variables. On the contrary, it is asymmetrically *influenced* by `Politics`.

5.2 Asymmetry affects intervention dynamics

We now investigate differences in belief system behaviour under intervention for symmetric and asymmetric models calibrated to the climate beliefs dataset. In this section we are concerned with *collective* impacts, e.g., the proportion of individuals whose behaviour shifts due to an intervention. This reflects population-level intervention scenarios, such as media campaigns or policies which affect most individuals. We will return to *individual* impacts of intervention in the following chapter (Chapter 6.1).

We consider both **outbound** and **inbound** interventions, illustrated below. Outbound intervention experiments (left) examine how interventions on a particular spin propagate to other beliefs and attitudes. On the other hand, inbound intervention experiments (right) consider a single belief or attitude as the intervention target and examine the differences in effects on this spin when intervening elsewhere in the network. We refer to the belief or attitude on which an intervention is applied as the **point-of-intervention** and the belief or attitude whose resulting state is measured as the **target**.



For outbound intervention experiments we examine the effects of intervening on the following spins:

- `Weather worry`: Level of worry about future extreme weather events
- `CC Worry`: Level of worry about current and future climate change
- `Politics`: General political party identification and ideology

These choices are motivated by our findings in the previous section. Recall that `CC Worry` and `Politics` both feature in the set of asymmetric relations, including in the relation between these two variables. `Weather worry` does not appear in any significant asymmetric relations, and—in comparison with the other variables—has relatively small outgoing interaction effects.

For inbound intervention experiments we consider interventions seeking to influence the `CC Action` variable, which captures individuals' general support for pro-environmental action on climate change (e.g., specific policies, support for increased regulation, and affirmative attitudes toward individual responsibility on climate change).

We define two quantities of interest: the **effect of intervention**, and **effect of asymmetry**. These are used, respectively, to assess an intervention's impact on a given spin's behaviour with reference to our expectations in a no-intervention scenario (Definition 5.2.1), and the difference in a spin's behaviour in the asymmetric model, compared to our expectations in the symmetric model (Definition 5.2.2). Note that the effect of asymmetry is not inherently concerned with intervention, but is a general measure for the difference between asymmetric and symmetry model dynamics.

Both quantities compute a difference in effects between two distinct models: the intervention and null models for the effect of intervention, and the asymmetric and symmetric models for the effect of asymmetry. To ensure outcome comparability, differences are computed between models with identical random number generation contexts.

Definition 5.2.1 (Effect of Intervention) Let $\mathcal{M}$ be a non-equilibrium belief system model, and denote the result of simulating the model for $t \in \mathbb{N}$ timesteps with initial state $\boldsymbol{s}_0$ as $\mathcal{M}^t(\boldsymbol{s}_0)$.

For an intervention model $\mathcal{M}_\delta$ with an arbitrary point-of-intervention, we define the **effect of intervention** for $\mathcal{M}_\delta$ as the change in outcome at time $t \in \mathbb{N}$, with respect to the null (no-intervention) model, $\mathcal{M}_0$:

$$f_{\text{int}}(\mathcal{M}_\delta) = \mathcal{M}_\delta^t(\boldsymbol{s}_0) - \mathcal{M}_0^t(\boldsymbol{s}_0) \tag{5.2}$$

Definition 5.2.2 (Effect of Asymmetry) Let $\mathcal{M}_{\text{asym}}$ be a calibrated asymmetric non-equilibrium belief system model with an arbitrary intervention applied, and $\mathcal{M}_{\text{sym}}$ be a corresponding symmetric model calibrated to the same dataset. We define the **effect of asymmetry** for $\mathcal{M}_{\text{asym}}$ as the difference in the effect of intervention with respect to the symmetric model:

$$f_{\text{asym}}(\mathcal{M}_{\text{asym}}) = f_{\text{int}}(\mathcal{M}_{\text{asym}}) - f_{\text{int}}(\mathcal{M}_{\text{sym}}) \tag{5.3}$$

In each of the intervention experiments described below, we use the models calibrated to the full climate beliefs dataset (Chapter 4). For a given experiment, we initialise the associated model (symmetric/asymmetric; intervention/null) with the binarised observations from the final wave of the climate beliefs dataset. We then draw consecutive samples from the model using parallel glauber dynamics (Chapter 2.3) until $t = 5$, before measuring the state of the target spin. Since the duration between waves in the climate beliefs dataset is approximately six months, the choice of $t = 5$ corresponds to measuring the impact of intervention after roughly two and a half years in the model timescale. To account for stochasticity in both the binarisation and simulation procedures we perform 500 repeats of each intervention.

We first consider the impact of intervention strength on effect of intervention. The strength of an intervention affects the degree to which the intervention changes the state of the point-of-intervention. However, this is also sensitive to the existing influence

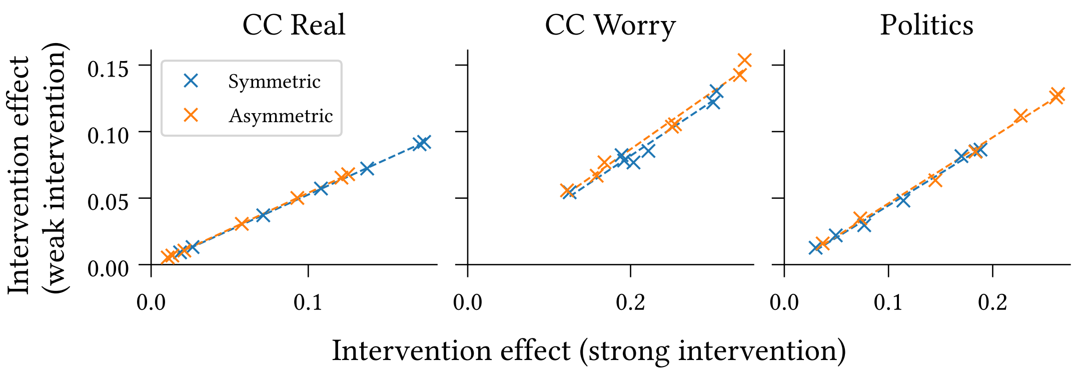


Figure 5.4: Weak ($\delta_h = 0.5$) and strong ($\delta_h = 2.5$) interventions exhibit strong positively correlated effects of intervention, consistently across points-of-intervention (panels) and intervention targets (scatterplots). This finding holds for both symmetric and asymmetric models calibrated to the climate beliefs dataset.

on that spin[9]. Since the climate beliefs dataset comprises individuals with different states, it follows that intervention strength may impact not only the degree to which the behaviour of a target spin changes, but also the relative effects of different interventions.

To understand the potential impacts of intervention strength, we compare the mean effect of intervention across individuals for intervention strengths $\delta_h \in \{0.5, 2.5\}$, for each of the outbound points of intervention listed above Figure 5.4. The choice of intervention strengths is motivated by our earlier analysis in Chapter 2.4, with the two strengths corresponding to 'weak' and 'strong' interventions respectively, as judged by their direct impact on the behaviour of the point-of-intervention.

We observe a strong linear relationship between intervention strength effects for each point-of-intervention, indicating that—at the population-mean level—the impact of intervention strength on effect is mostly one of scale, that applies similarly to all target spins. For the purposes of the following experiments it then suffices to consider only a single intervention strength. As our current focus is on intervention propagation and the effects felt at spins other than the point-of-intervention, we take $\delta_h = 2.5$, such that the interventions can be considered generally effective at shifting the behaviour at the point-of-intervention.

Figures 5.5 and 5.6 show the outbound mean effects of intervention and asymmetry, respectively, across individuals, for the points of intervention listed above. The confidence intervals show two standard deviations around the mean values.

We observe, in Figure 5.5, that different targets exhibit different effects of intervention. The effect generally decreases with the magnitude of the interaction from the point-

[9]For detailed discussion on the selection and implications of intervention strengths see Chapter 2.4.

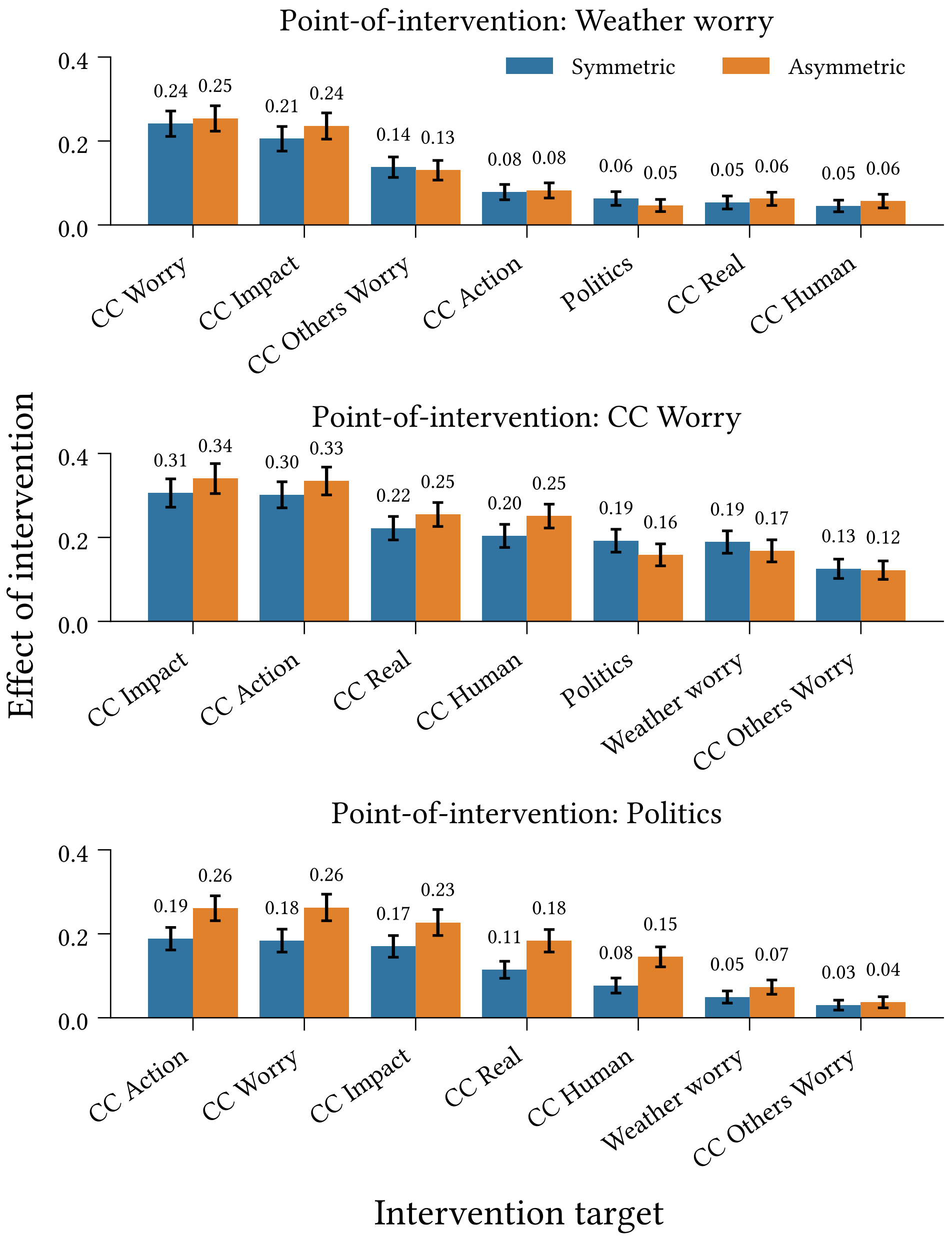


Figure 5.5: Outbound effect of intervention (Definition 5.2.1) at each target spin for interventions targeting `Weather Worry`, `CC Worry`, and `Politics`, in symmetric and asymmetric belief system models calibrated to the climate beliefs dataset. Intervention strength:w $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect, measured across 500 repeated simulations.

of-intervention to the target spin; however, we also note that all effects are positive. This suggests that for the measurement time examined here, interventions act primarily through direct interactions with the target spin, with some indirect influence propagating through intermediary interactions. For instance, while `Weather Worry` has no direct influence on `CC Action`, we see nonzero effect of intervention as a result of indirect paths, e.g., through `CC Worry`. We see further evidence of this through comparison with

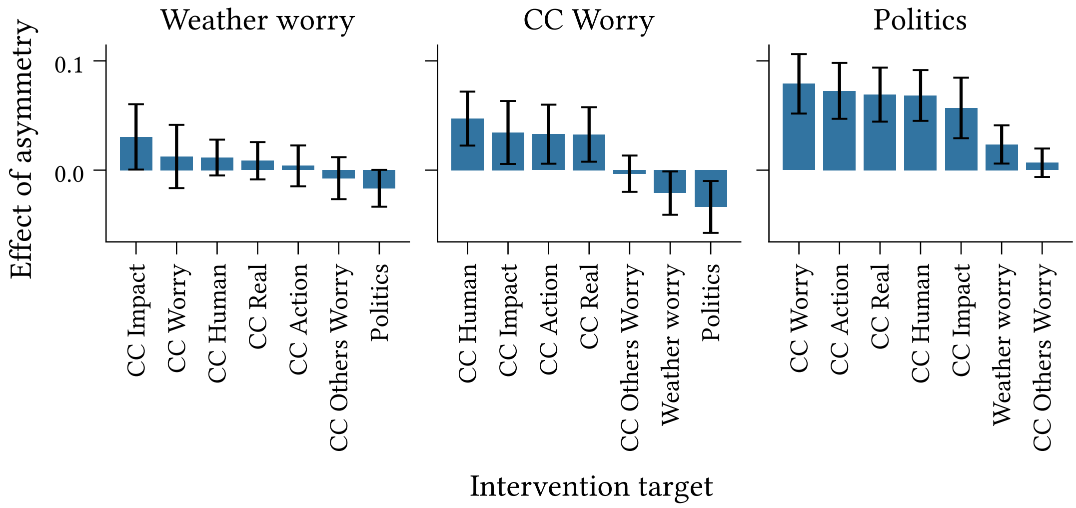


Figure 5.6: Outbound effect of asymmetry (Definition 5.2.2) at each target spin for interventions on `Weather Worry`, `CC Worry`, and `Politics`, for models calibrated to the climate beliefs dataset. Intervention strength: $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect of asymmetry ($n = 500$).

an analogous figure (see Figure B.1 in Appendix) using a measurement time of $t = 10$ (approximate five years in the models' timescales), which shows larger increases in the effect of intervention for variables with stronger incoming interactions from non-intervention spins.

While the confidence intervals for the symmetric and asymmetric models overlap significantly for most targets in Figure 5.5 (with some exception in the bottom panel), the effects of asymmetry displayed in Figure 5.6 provide a cleaner comparison. We observe no significant differences between the two models for the `Weather Worry` scenario. For the remaining two scenarios, however, we see several significant differences. In particular, all pairs of variables with asymmetric direct relations—identified in the previous section—display significant differences.

Comparing `Politics` and `CC Worry`, we observe that while `CC Worry` exhibits greater effects of intervention (Figure 5.5), `Politics` exhibits larger mean effects of asymmetry (Figure 5.6), though the latter differences are not statistically significant. In both cases we see that in the symmetric model the strengths of cross-interactions with other spins are typically between the corresponding inbound and outbound asymmetric interaction strengths. We contrast this with `CC Impact`, which has mostly symmetric relations. While `CC Impact` and `Politics` have extremely similar outbound interaction strengths in the asymmetric model, `CC Impact` is considerably more influential than `Politics` in the symmetric model (i.e., it has higher average interaction effects).

Moreover, while `CC Worry` has inbound and outbound interactions with all other spins in both models, `Politics` has different inbound and outbound connectivity in the asymmetric model. We see this reflected in the symmetric model, which omits interactions with `CC Human` and `Weather Worry` entirely despite, outbound interactions to these spins in the asymmetric model. This explains the observed differences in the effects of intervention and asymmetry: `CC Worry` has higher effects of intervention in both models on account of its strong outbound interaction effects, which are also mostly retained in the symmetric model, while the removal of interactions for `Politics` results in lower influence and fewer (direct and indirect) paths in the symmetric model, compared with the asymmetric model.

Turning our attention toward inbound intervention dynamics, we now consider the effects of different interventions seeking to influence the state of `CC Action`. The top panel of Figure 5.7 shows the mean effect of intervention observed at the target spin for each possible point-of-intervention. Examining the mean effects allows us to gauge the expected absolute effect of each intervention, but is limited for purposes of comparing the relative effects of interventions.

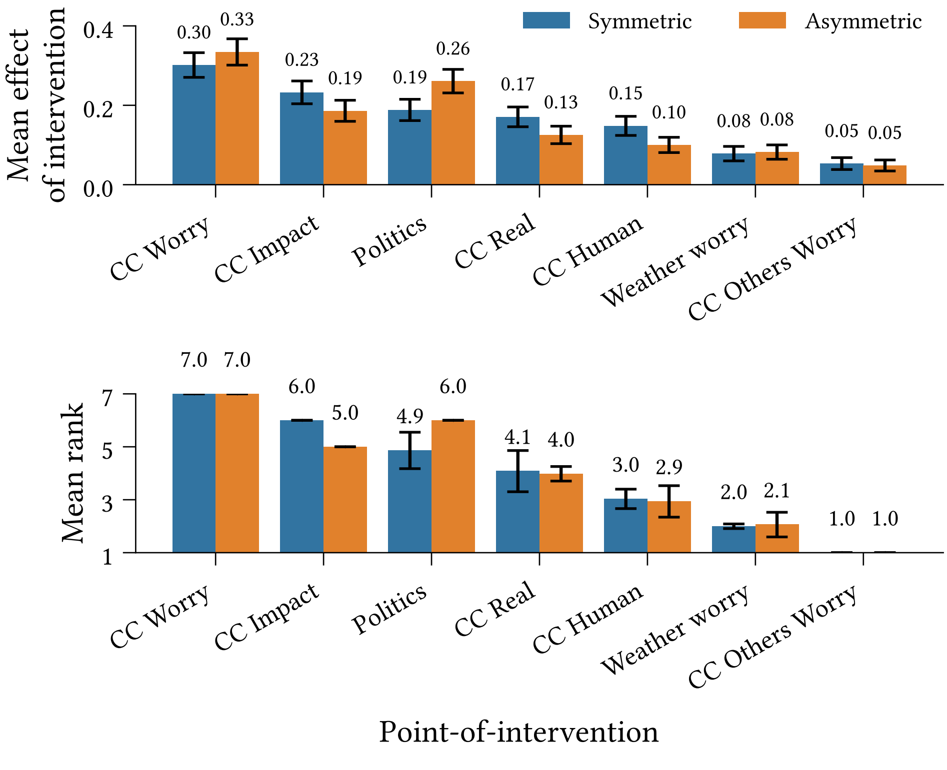


Figure 5.7: Inbound mean effect of intervention (Definition 5.2.1) (*Top*) and mean rank (*Bottom*) for interventions targeting `CC Action`. Ranks are calculated per-repeat with respect to the average effect of intervention—higher values are assigned higher ranks. Confidence intervals display 1.96 standard deviations ($n = 500$).

The mean effect of intervention, calculated separately for each point-of-intervention, erases information regarding the expected relative effectiveness of different interventions. To assess relative effectiveness, we therefore also estimate the expected ranking over points-of-intervention (the bottom panel of Figure 5.7). For each repeat, we order the points-of-intervention in increasing order of average collective effect of intervention. Higher values are assigned higher ranks. In the case of a tie, we assign all tied spins the minimum rank which would have been assigned to the group, had they been distinct[10].

As with the outbound case, we find that the effect of intervention on `CC Action` varies depending on where we intervene. Points-of-intervention (along the x-axes) are sorted in decreasing order of mean values. We observe identical orderings between the effect of intervention and ranking plots, showing good correspondence between the two measures. We observe a clear inversion in this ordering for the asymmetric model, with `Politics` exhibiting the second-highest rank, and (at least) second-highest mean effect of intervention ($p < 0.05$).

Unlike with the outbound effects, the order of effect sizes does not align well with the corresponding strengths of direct interactions strengths into the target variable. For instance, `CC Real` and `CC Human` have the second- and third-strongest cross-interactions into `CC Action` in the symmetric model, yet the fourth- and fifth-highest effect of intervention. This likely reflects the low outbound connectivity of these variables in comparison with `CC Impact` and `Politics`, despite the latter spins having relatively lower direct on the target—additional evidence that indirect influence plays a critical role in intervention dynamics.

Most confidence intervals around the mean ranks are small, indicating negligible (if any) variation between simulations or possible binarisations of the dataset. However, we do observe three larger, overlapping confidence intervals for each model. These arise due to variation the mean effect of intervention causing occasional inversions of the ranking order. Note that we only observe inversions between points-of-intervention which are adjacent in the sorted order.

[10]For instance, suppose we have five variables, $\{A, B, C, D, E\}$ with average collective effects $\{0.2, 0.2, 0.6, 0.4, 0.1\}$. Then the assigned ranking would be $\mathrm{Rank}(A, B, C, D, E) = [2, 2, 5, 4, 1]$. The variables B and C are both assigned the rank 2. No variable is assigned rank 3.

Chapter 6

Heterogeneous belief systems and intervention effects

In this chapter we investigate individual heterogeneity in both intervention behaviour and belief system structure in asymmetric, non-equilibrium belief system models.

While the previous chapter was primarily concerned with the *collective* effects of intervention, the distribution of expected effect of intervention across individuals for inbound interventions targeting 'Climate Action' shows substantial variation between survey participants (Figure 6.1). We begin this chapter with an investigation into the conditions under which different interventions targeting attitudes on climate action are likely to be effective.

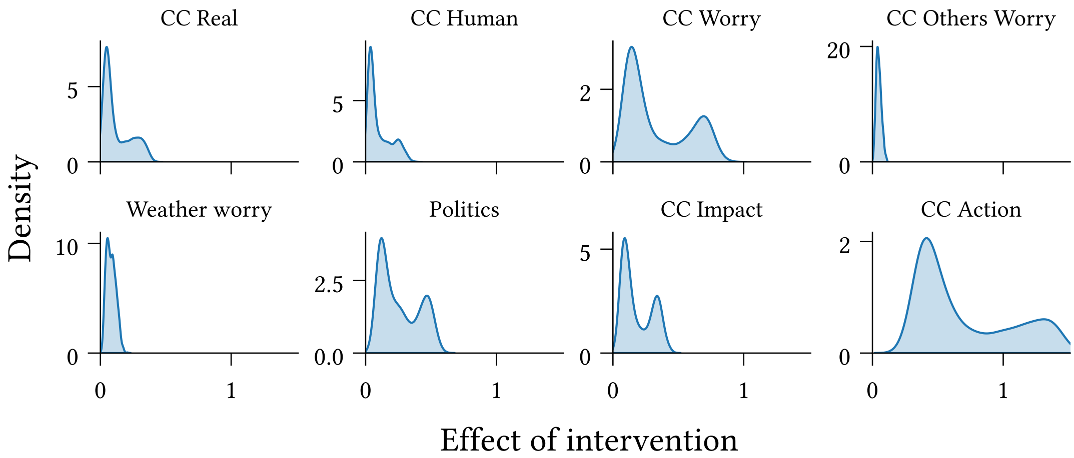


Figure 6.1: The expected effect of intervention for strong interventions ($\delta_h = 2.5$) targeting 'Climate Action' exhibits substantial variation between individuals.

We then consider how inferred belief system structure varies between individuals, or subgroups in a population. In the previous chapter we identified political ideology as a particularly influential attitude dimension, both through asymmetric interactions with other variables (Chapter 5.1) and as an effective point-of-intervention for interventions targeting attitudes on climate action (Chapter 5.2). Here, we investigate whether this influence extends beyond pairwise interactions with other variables by comparing asymmetric belief systems calibrated to the (self-reported) liberal and conservative subpopulations within the climate beliefs dataset.

6.1 Heterogeneity in intervention effects

In the intervention simulations described in the previous chapter, the only distinguishing factor between survey participants is their binarised initial state derived from the final wave of the climate beliefs dataset. It follows that any difference in belief system dynamics between distinct individuals—after accounting for simulation stochasticity—is the result of their different initial states. To identify the conditions under which an intervention is effective, it then suffices to characterise the set of *initial states* which yield effective interventions, and distinguish them from those that do not.

Our goal is to find a function which maps from an intervention effect to sets of (unbinarised) states that yield that effect, $g : \mathbb{R} \to 2^X$, where $X = [-1, +1]^N \subset \mathbb{R}^N$. We'll call g the **effect characterisation function**.

Consider a function which does the opposite, mapping initial states to a measure of intervention effect, $f : X \to \mathbb{R}$. Given such a function, we can straightforwardly construct the corresponding effect characterisation function as:

$$g : y \mapsto \left\{x \in \pm 1^N \subset \mathbb{R}^N \mid f(x) = y\right\} \tag{6.1}$$

However, while technically satisfying the definition of the effect characterisation function, such a function would be useless for qualitatively determining the kinds of initial states which yield effective interventions. The elements of the codomain have (potentially) infinite cardinality, and are not immediately interpretable. For our purposes, we are less interested in the *specific* states that yield a given intervention effect, but more so in a *concise description* of that set of states.

Regression decision tree models can provide such descriptions, using inequality bounds to partition the initial state space into regions, each of which is assigned a predicted effect. Given a parameterised regression decision tree, we may construct a concise effect

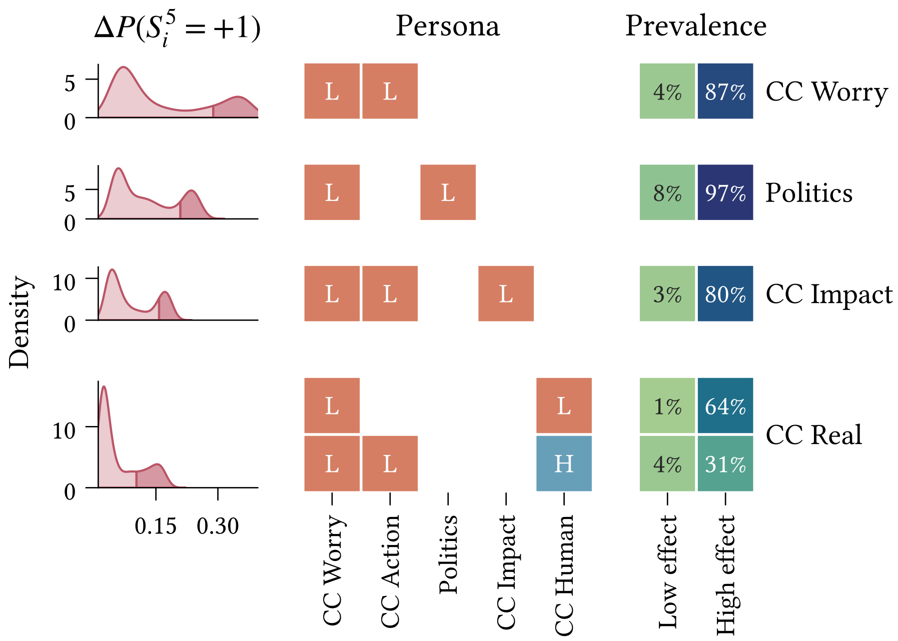


Figure 6.2: Characterisation of initial states ('personas'; middle panel) which yield effective interventions when targeting `CC Action` using different points-of-intervention (rows). Intervention effect is measured as the difference in `CC Action` activation probability between intervention ($\delta_h = 2.5$) and null (no-intervention) models at $t = 5$, for the asymmetric belief model calibrated to the climate beliefs dataset. Differences in the population upper quartile are 'high-effect'. (**Left**) Distribution of intervention effect across survey participants. (**Centre**) Typical high-effect personas estimated using a depth-3 regression decision tree. Rows are distinct personas. Cells correspond to intervals of initial state dimensions ($L = [-1, 0]$, $H = [0, +1]$). (**Right**) Prevalence of each persona in the high-effect and low-effect subpopulations.

characterisation function by identifying, for each predicted effect, the combination of inequalities which define the corresponding infinite set of initial states. When the parameter estimation algorithm is restricted to shallow trees (e.g., depth 3 or 4, referring to the number of inequality bounds defining each region), these combinations can also be interpretable as rules or *personas*.

Figure 6.2 shows the effect characterisation functions for different interventions targeting `CC Action`, obtained from shallow regression decision trees with depth 3. The response variable is the expected difference between the activation probabilities for `CC Action` at $t = 5$, in the intervention and null models, estimated per-individual using the average probabilities across 500 repeated simulations. Each model is calibrated to the full climate beliefs dataset (see Chapter 4 for calibration details). As in the previous chapter, we use a fixed intervention strength of $\delta_h = 2.5$. The decision trees are fit using a mean-squared error objective function, and each achieve R^2 values of at least 0.9 (estimated using 10-fold cross-validation).

Each row in the figure corresponds to a different point-of-intervention (specified on the right). The left-hand column shows the distribution of differences in activation probabilities across individuals. The central column shows the personas which yield intervention effects within the upper quartile. Each row is a separate persona, and the columns refer to different dimensions (beliefs/attitudes) of the initial state. The personas themselves are specified using a compressed representation[11]: L $\mapsto [-1, 0]$ and H $\mapsto [0, +1]$. The right-hand column displays the prevalence of each persona within/ outside the upper quartile of individuals, showing the proportional make-up of the high-effect population. We exclude points-of-intervention with no sufficiently high-effect observations (e.g., `CC Others Worry` in Figure 6.1), classified as an upper quartile less than 0.1 (`CC Human`, `CC Others Worry`, `Weather Worry`).

Since the regression decision tree produces a full tree, all personas have size 3 by default. However, these can often be compressed. When two personas differ only along one feature dimension, split at the same value, and both predict high-effect interventions, we combine them into a single persona which omits that feature (i.e., spans the entire feature dimension). This is observed, for instance, in the persona for interventions on `CC Worry`.

The expected difference in activation probabilities between the intervention and null models exhibits a clear bimodal distribution across survey participants for each scenario, with the higher mode contained within the upper quartile (indicated by the darker shaded regions). For each point-of-intervention we observe a small set of personas. In each case these exhibit high prevelence among individuals with high intervention effects, and considerably lower prevalence for other individuals, indicating that the identified personas effectively characterise the conditions for effective interventions[12] . While prevalence among the high-effect individuals is generally high, this does vary across points-of-intervention (e.g., 20% of effective interventions on beliefs about climate impacts are not captured).

For the present analysis, it is important to recognise that the intervention effect measured here does not, in general, reflect the resulting change in behaviour with respect to the initial state. It is entirely possible that both intervention and null models lead to an increase, or decrease, in the probability of desired behaviour for the target spin.

[11] As our intention is to capture coarse patterns, we round the original `f64` values describing the interval boundaries extracted from the model to the nearest multiple of 0.5.

[12] Note that the personas are not necessarily complete. For instance, suppose that a pair of variables are highly correlated in the initial state, and are Low whenever the intervention effect is high. A complete characterisation includes both variables; however, a decision tree is likely to include only one, since after splitting on one of the two variables, the other is redundant. This is especially true for shallow decision trees.

Hence it is most appropriate to view 'high-effect' interventions as those which achieve substantially more desirable states than would be observed given no intervention.

With this in mind, we now consider several specific features of interest in the identified personas. Firstly, we observe that all personas require a low initial state for `CC Worry`. This may result from this variable's low inertia and high connectivity — in particular, its large outbound interaction effect toward the target variable (see Figure 5.2). These factors result in `CC Worry` being relatively influential and influentia*ble*, and therefore an effective indirect pathway for various interventions targeting `CC Action`. The significance of the requirement that `CC Worry` be *low* is evident when comparing the implications for the null and intervention models. Due to `CC Worry`'s considerable outbound interactions, an intervention which successfully activates this variable has increased potential for propagation. In the null model, however, these interactions work against the desired result—if `CC Worry` remains low, it exerts this influence on all other spins.

Second, we observe that for each point-of-intervention (with the exception of `CC Real` on account of high correlation with `CC Human`[12]), a necessary condition for high effect is that the initial state of point of intervention itself be low. That is, for an intervention on X to be successful, X must not already be too high. This aligns with our prior expectations regarding the varied effects of interventions with respect to pre-intervention state (see Chapter 2.4).

Finally, the `CC Real` scenario is unique in that it includes two prevalent personas. The more prevalent persona corresponds to situations in which individuals are concerned about the current/future effects of climate change, but do not believe that climate change is human-caused[13]. In the case where an individual does believe that climate change is human-caused, this intervention may still be effective, so long as they do not already have a positive attitude toward climate action.

6.2 Heterogeneity in belief system structure

To-do:

- Discuss differences in baseline activations.

Until this point, we have considered belief systems as common to a population of individuals. However, the relations between beliefs and attitudes are inherently individual

[13]The negative state for `CC Human` is an aggregation of the beliefs that climate change is a natural phenomenon, and that climate change is not real (has no causes). See Chapter 9 for further details.

in nature. The existence, direction, and degree of relation between two beliefs is dependent on an individual's own beliefs regarding their relatedness.

Here we investigate the extent to which belief systems may vary between groups of individuals with different self-reported political ideologies. While this is still far from representative of the differences between individuals (Brandt & Morgan, 2022), it will allow us to examine differences in general trends for these subpopulations. Using the parameter estimation approach described in Chapters 3.3 and Chapter 4, we calibrate separate asymmetric belief systems to the subsets of the climate beliefs dataset which consistently (i.e., in both waves) self-report conservative ($n = 507$) and liberal ($n = 375$) political ideologies[14]. We exclude the `Politics` variable from the model, since this is captured by the partitioned datasets. The hyperparameters used for regularisation strength and smoothing are listed in Table 3.1.

Figures related to model calibration can be found in Appendix B. The estimated parameters exhibit considerably higher uncertainty than in the model calibrated to the complete dataset due to the smaller sample sizes (Figure B.2). We find only one significant case of asymmetry, on `CC Worry` $\longrightarrow$ `CC Impact` in the conservative model (Figure B.3), and only a minority of significant differences ($p < 0.05$) between interaction strengths, all for edges which are stronger in the conservative model (Figure B.4).

Figure 6.3 shows the interaction effect matrices, $\boldsymbol{J}$, for the conservative and liberal subpopulations. We observe several differences between the two models, as well as with comparison to the model calibrated on the full dataset (Figure 5.2). Notably, the model calibrated to the liberal subpopulation has higher sparsity (proportion of missing cross-interaction edges) and mean interaction effect over cross-interactions than either the conservative model or the complete model (Table 6.1). Among the interactions present in the liberal model, 82% also occur in the conservative model, contrasting the 54% of conservative interactions which are also in the liberal model.

The conservative model displays broad (yet mostly weak) reinforcing interactions between climate-related concerns, beliefs and others' concerns, and climate-impact beliefs. This contrasts the liberal model, in which only `CC Worry` and `CC Impact` are non-trivially related. Moreover, we observe that `CC Others Worry` and `Weather Worry` in fact have *no* incoming cross-interactions.

We also note differences in the interaction effects influencing `CC Action` between the two models. While the liberal model expects attitudes toward climate action to be

[14]Note that the total number of samples across the conservative and liberal subsets is smaller than the number of samples in the complete climate beliefs dataset, since we only retain data from participants whose ideology is with consistent across survey waves.

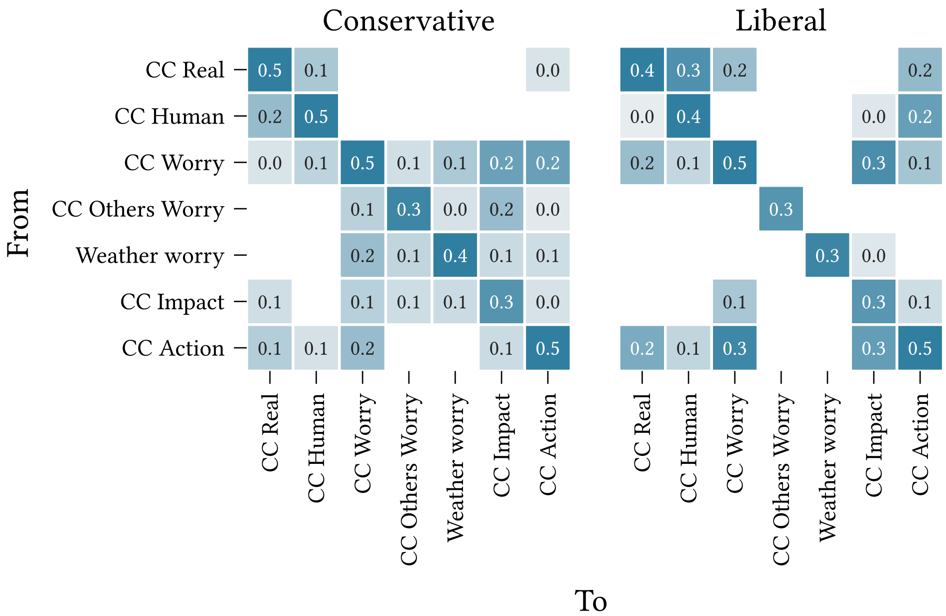


Figure 6.3: Interaction effect matrices ($\boldsymbol{J}$) for asymmetric belief systems calibrated to the conservative and liberal subsets of the climate beliefs dataset (Chapter 9) using the parameter estimation method in Chapter 3.3.

influenced substantially by individuals' beliefs regarding the existence and nature of climate change, these influences are absent or trivial in the conservative model. Instead, we observe that concern about climate change (`CC Worry`) is the only large cross-interaction toward `CC Action` in the conservative model.

Comparing now with the complete model (calibrated to the full dataset), we find that interactions absent from the complete model are generally also absent from both smaller models. The exceptions are {`CC Others Worry`, `Weather Worry`} $\longrightarrow$ `CC Action`, which are present in the conservative model, albeit with small effect sizes.

The converse does not hold; in several cases edges are absent from the smaller models, yet included in the complete model (e.g., `CC Human` $\longrightarrow$ `CC Action` in the conservative model, `CC Worry` $\longrightarrow$ `Weather Worry` in the liberal model). While it is tempting to interpret these

Data subset	Sparsity	Mean interaction
Full	0.30	0.13
Conservative	0.38	0.10
Liberal	0.60	0.17

Table 6.1: Sparsity (proportion of missing cross-interactions) and mean interaction effect (over cross-interactions) for asymmetric models calibrated to the conservative and liberal subsets of the climate beliefs dataset, compared with the model calibrated on the complete dataset.

as differences in the ideology-specific belief systems, this inference is not necessarily valid. Given the smaller datasets used to calibrate these models—and the slow-moving dynamics of the measured variables—these edges may be excluded on the basis that we do not observe their effects. This is less likely for effects which are more substantial in the complete model, and thus supported better by the dataset.

Chapter 7

Discussion

In this chapter we review the research questions posed back in Chapter 1 (restated below) in light of the results presented in the previous two chapters, and discuss their place in the broader context of belief system dynamics. We then conclude the chapter with a discussion on the limitations of our findings and implications for future work on belief system dynamics and belief-level interventions.

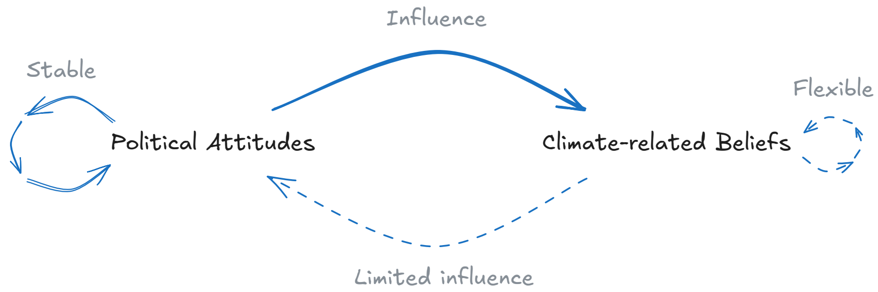
RQ1: *To what extent are belief-level influences* symmetric *or* asymmetric*, in models of climate change belief systems inferred from the climate beliefs dataset?*

On the question of existence, Chapter 5.1 demonstrated asymmetric relations between several pairs of beliefs and attitudes, while also finding that asymmetry may not *always* be exhibited. Two attitudes—political ideology/alignment and climate-related concerns—displayed significant excess influence over several other beliefs and attitudes.

How should we interpret these asymmetric relations? Recall that relations in the non-equilibrium belief system model reflect temporal influence—how much one belief or attitude constrains[15] the future state of another (Chapter 2). An asymmetric relation reflects a constraint differential, where one belief or attitude has greater influence on the other than vice versa. More generally, the asymmetric model distinguishes between two forces: influence and influentiability. Influence (the strength of outbound interactions) determines how the extent to which one belief or attitude constrains others, while influentiability (the strength of inbound interactions) determines the extent to which its own behaviour is constrained by others.

[15]Note that this interpretation (in particular, the use of the verb 'constrains') depends on the assumption that the calibrated model captures within-person associations. We discuss this matter in some depth later in this chapter.

Political identity being more influential than influentiable in this context is consistent with previous studies on its mutual influences with climate-related beliefs and attitudes in the USA. First, political identity has been shown to substantially impact both climate-related beliefs (Whitmarsh, 2011), as well as support for climate policies (Bumann, 2021), with individuals tending to support or oppose specific policies on the basis of partisan identification rather than policy content (Unsworth & Fielding, 2014; Van Boven et al., 2018).



At the same time, while public beliefs and attitudes toward climate change in the USA have shifted significantly over the past two decades (Hamilton et al., 2015; Marlon et al., 2022), political identity has been shown to be highly stable over similar timeframes (Brandt & Morgan, 2022; Green & Platzman, 2024). This suggests that while political identity contributes meaningfully to climate-related beliefs and attitudes, changes in these states are less likely to incite changes in political identity.

Climate-related concern exhibits asymmetric influence over beliefs regarding the anthropogenic nature and current impacts of climate change. Both relationships have received comparatively less attention in prior research than those of political attitudes with climate beliefs and attitudes. One possible explanation for these asymmetric relations is that climate worry and climate beliefs influence one another through different mechanisms. For instance, Mead et al. (2012) suggest that climate worry may promote information-seeking behaviour, leading to changes in climate-related beliefs.

For both variables, we found significant differences between inbound and outbound centrality indices (strength centrality for climate worry; strength and degree centrality for political ideology/alignment). The undirected-network versions of these centrality indices are often used to assess belief importance, influence, or position in belief systems. Our findings indicate that such assessments may be misleading for beliefs or attitudes involved in several asymmetric relations.

RQ2: *How do asymmetric and symmetric beliefs systems differ with regards to intervention strategy and effectiveness, in models inferred from the climate beliefs dataset?*

The consequences of asymmetry for belief system dynamics were subsequently reflected in Chapter 5.2, where interventions on political ideology/alignment in the asymmetric model were found to be almost universally more effective than in the symmetric model. Moreover, for interventions targeting attitudes toward climate action, political ideology/alignment ranked higher with regards to collective effect in the asymmetric model than the symmetric model, displacing a mostly-symmetric belief with similar outbound interactions. Interestingly, while interventions on climate-related concerns typically outperformed those on political ideology/alignment in terms of absolute effect of intervention, the differences between the asymmetric and symmetric models for this variable were comparatively less pronounced.

It is important to note that the effectiveness of interventions on political ideology/alignment is *despite* this variable's high inertia (i.e., strong self-interaction effect). When this variable is negative, high inertia leads to a lower pre-intervention effective baseline activation, making it harder to intervene. The effectiveness of interventions on this variable may be explained by fact that if an intervention successfully 'flips' political ideology/alignment, the high inertia works in favour of the intervention.

For the measurement timescale used in these experiments (approximately 2.5 years), interventions appeared to act primarily through direct interactions, though smaller indirect effects were also present; this is consistent with empirical results obtained by Chambon et al. (2022) in an experimental study concerning beliefs and attitudes relating to COVID-19. Additional experiments included in Appendix B found that longer timeframes led to increased indirect intervention effects. The presence of indirect effects is broadly expected given the pre-defined model dynamics; however, the observed magnitudes suggest that while state changes may propagate beyond direct connections, this process is typically slow. We return to this point shortly in our discussion of **RQ3** (↪), which finds that some indirect propagation may nonetheless be instrumental to achieving effective interventions at an individual level.

This distinction in behaviour between political ideology/alignment and climate-related concern likely relates to differences in influentiality and influentiability between the two attitudes. Climate-related concern is more influential than political ideology/alignment, on account of its increased outbound interactions and interaction strength. The key difference, however, lies in the attitudes' influentiability; while climate-related concern has inbound and outbound interactions with all other beliefs and attitudes, political ideology/alignment has considerably fewer *inbound* than *outbound* edges[16].

[16]Note that this point pertains to the model calibrated on the complete climate beliefs dataset. This model omits several interactions (due to regularisation) which are inconsistently excluded from the bootstrap models used to assess the significance of asymmetric relations.

That is, several beliefs and attitudes influenced by this variable have no influence on it themselves. Since the symmetric model must include or exclude both directional interactions between a pair of beliefs, cases where the asymmetric model specifies an interaction in only one direction necessarily lead to differences in model behaviour. This draws attention to a broader issue regarding relational symmetry assumptions.

One of the symmetric model's key strengths is its small(er) parameter count. With fewer degrees of freedom than the asymmetric model, it can achieve more accurate parameter estimates when calibrated to the same dataset (i.e., smaller confidence intervals in Figure 4.3). However, this comes at the cost of model misspecification when a subset of the true relations are asymmetric. We might expect the inferred symmetric influence relations to be similar to the average of the corresponding directed effects, yet this reasoning turns out to be flawed for at least two reasons. Firstly, interaction effects are not estimated in isolation, but jointly with all other parameters which influencing the same spin (or *both* spins in the symmetric case). Hence, when one interaction parameter changes—for instance, if we replace a pair of asymmetric interactions by their average—the rest are likely to change as well. Secondly, regularisation is often used to obtain sparse network representations and reduce overfitting, and distorts parameter values non-linearly in the process. In reality we find that while most symmetric interactions lie between their asymmetric analogues, this is not always the case, and those that are do not fall predictably near the middle. As observed in the case of political ideology/alignment, the symmetric model may also exclude pairwise interactions altogether, or create a (bi-)directional interaction where influence actually flows unidirectionally.

RQ3: *How do intervention outcome and effectiveness vary between individuals with different initial conditions in asymmetric belief systems inferred from the climate beliefs dataset?*

In Chapter 6.1 we found that intervention effectiveness depends predictably on individuals' pre-intervention belief states. Most high-effect interventions targeting attitudes toward climate action required low initial values for both the point-of-intervention and target.

Surprisingly, pre-intervention climate-related concern was required to be low for *all* of the personas identified as characteristic of effective interventions, even when this was neither the point-of-intervention, nor the target. This finding may be explained by the combination of high influence and influentiability associated with climate-related concern, thereby making this variable an effective indirect route for various interventions (not restricted to this particular target). This stands in contrast with political ideology/alignment, which has similarly high influence on the target variable, but is harder to influence, as discussed above. At first glance this finding appears to contradict our earlier discussion on **RQ2** (↩), which found indirect interventions to have limited impact on collective effects over short timeframes. However, this rather reflects the diversity in intervention outcomes across individuals.

In general, the identified personas characterise the conditions for effective interventions accurately. The rare instances where individuals with these traits fell outside the high-effect region may be attributable to the rough-and-ready use of the upper quartile to classify high-effect interventions. On the other hand, while the personas capture *most* cases where interventions are effective, this varies between points-of-intervention. This directly reflects the limited representational capacity of effect characterisation functions based on shallow decision trees; a tree depth of three permits at most eight personas[17] to describe the full range of intervention effectiveness, each of which comprises at most three conditions.

The results also highlighted a separate issue with our regression decision tree approach to characterising intervention effectiveness, namely that the descriptions may be incomplete when important features are highly correlated. This is a result of the decision tree optimisation procedure, which selects the 'splits' which best account for unexplained variance. We saw this reflected in the characterisation of effects for interventions on beliefs about the existence of climate change. While effective interventions here generally require that the initial point-of-intervention state be low, this was omitted due to high correlation between this variable and beliefs regarding the causes of climate change.

As a final point, the findings from this experiment must be interpreted in the context of the belief system model calibrated to the complete dataset. Individual differences in belief systems (as discussed in a moment) are likely to result in heterogeneous responses to interventions beyond those characterised here. Potentially influential factors include political identity and social norms. Both vary individually (Laursen & Faur, 2022) and socially (Brown & Enos, 2021; Laursen & Faur, 2022; Webster et al., 2022), and both have been demonstrated as moderating the effects of interventions in the USA targeting support for climate policy and related attitudes (Allcott, 2011; Gromet et al., 2013; Unsworth & Fielding, 2014; van Valkengoed et al., 2022).

RQ4: *How do asymmetric belief systems inferred from the climate beliefs dataset vary between conservative and liberal individuals?*

[17]The personas are given by the paths from the root to each leaf; a binary tree with depth three has $2^3 = 8$ leaves, and thus eight personas.

Finally, Chapter 6.2 showed (potentially) substantive differences between asymmetric belief system models calibrated separately to conservative and liberal subpopulations; however, given the smaller sample sizes used to calibrate these models—and consequentially, higher parameter uncertainty—these results must be interpreted with caution.

When calibrated on the non-bootstrapped data subsets, each model featured high-magnitude interactions not present in the other, yet while the liberal model was considerably sparser than the conservative model, most interactions in the former were also present in the latter. In comparison with the asymmetric model calibrated on the complete dataset, used in the prior experiments, the ideological models were both sparser, yet together had high overlap in connectivity with the complete model. We found only one case of significant asymmetry, which was also present in the model fit to the complete dataset.

These findings demonstrate both significant differences and similarities between relational belief system structures, conditional on political ideology, in line with earlier studies comparing belief systems on partisan identity (Lee et al., 2024). The observed differences suggest the presence of higher-order relationships between beliefs and attitudes, consistent with the view that belief system relations, themselves, correspond to beliefs or attitudes (e.g., belief that two states of affairs are related) (Fishbein & Ajzen, 1977, p. 219).

Sparse networks imply a restricted set of commonly-observed transitions. This is, to an extent, expected for the liberal model, which is calibrated using fewer observations than the conservative model; however, the higher sparsity is also corroborated by significantly lower edge selection for several interaction in the liberal model across bootstrap samples. This finding may suggest a greater degree of consistency in the observed dynamics between individuals who self-identify as liberal.

Previous studies by Gregersen et al. (2020) and Lind et al. (2024) identified beliefs about the anthropogenic causes of climate change as more strongly associated with various climate-related beliefs and attitudes for non-right-leaning individuals. These findings were not replicated in our results; instead we found limited association (inbound or outbound) with this variable in both models. There are several possible explanations for this apparently contrary result, including parameter uncertainty in the present study. Firstly, both studies are based in a European, as opposed to US, context; Lee et al. (2025) showed cross-national differences in climate belief systems including related variables. Second, both measure association using cross-sectional correlational measures, which may differ considerably from the time-lagged interaction parameters used in the present study.

7.1 Limitations

When beliefs and attitudes are fairly stable—as in the climate beliefs dataset—cross-sectional methods for inferring belief system structure tend to identify *between*-person associations more so than the *within*-person associations which are typically desired (Brandt & Morgan, 2022). We partially mitigate this problem by calibrating to longitudinal data, and using self-interaction terms to capture persistence in belief/attitude states. However, our model does not account for belief/attitude stability due to the presence of stable traits (Hamaker et al., 2015; Usami et al., 2019).

Hamaker et al. suggest modelling individual random intercepts (in our case, individual baseline activations) which are fixed across waves and account for some of the effects of stable traits. However, at least three waves of data are required for the random intercept model to be identifiable[18], while the climate beliefs dataset comprises only two waves.

In unreported experiments we tried a similar approach, modelling baseline activations as linear functions of demographic factors (e.g., age, education, rural/urban status), which requires only two waves to be identifiable. While the resulting baseline activations varied substantially between individuals, indicating that they captured some differences in stable traits, we found little-to-no impact on either the inferred interactions or the intervention experiment results. Hence further investigation is required to determine the extent to which the models calibrated in Chapter 4 capture within-person associations.

Relatedly, since the climate beliefs dataset comprises only two waves, we cannot distinguish between belief system dynamics arising due to endogenous factors (the states of other beliefs and attitudes) and exogenous factors (current events which affect beliefs and attitudes). We implicitly assume that (i) exogenously-driven dynamics at an individual level are not substantially correlated between individuals, and thus not reflected in the model, and (ii) the time between measurements (approximately six months) is sufficiently short that widespread exogenous factors are negligible.

The second assumption has questionable validity, since for instance, both the 2020 US presidential election and January 6 Capitol attack occurred during this timeframe. However, we observed highly stable political attitudes over this period, consistent with prior studies (Green & Platzman, 2024). Given the significance and considerable media

[18]For a given individual, each pair of consecutive waves constitutes a single observation (given we model conditional transition probability). For a set of $N \in \mathbb{N}$ beliefs/attitudes the asymmetric model comprises $N^2 + N$ parameters, with M additional parameters when modelling individual baselines. Hence for $M \in \mathbb{N}$ individuals we require at least three observations per-individual for the problem to be determined. Note that more waves may be required when N is large, such that the number of non-intercept model parameters is larger than M.

coverage of both events, as well as their highly-political nature, we would expect political attitudes to be affected more so than climate-related beliefs/attitudes. The fact that we do not see this reflected may suggest minimal exogenous impacts, more generally, on the beliefs and attitudes considered in this study.

Our present focus on endogenous dynamics also does not discount the substantial role of social interaction in belief and attitude change (Converse, 1964; DeGroot, 1974; Galesic et al., 2021; Karashiali et al., 2023) or stability (Brown & Enos, 2021; Prentice & Miller, 1996), including in intervention contexts (Brewer et al., 2017). Rather, this decision reflects the fact that the dataset used in this study did not include social network information. The matter of integrating social and cognitive forces in similar belief system models has been explored in several accounts (Aiyappa et al., 2024; Dalege et al., 2025; Rodriguez et al., 2016; van der Maas et al., 2020), generally assuming individuals have dual objectives of cognitive consistency and social coherence (Festinger, 1962; Gawronski & Strack, 2012; Heider, 1946). The Networks of Belief theory, proposed by Dalege et al. (2025), builds on the Causal Attitude Network model which also serves as the foundation for our proposed model. Since our work is mostly orthogonal to theirs, we anticipate that it would be straightforward to incorporate such social influences in our model.

We now discuss three representational limitations of the model used in the present study, pertaining to (i) the use of pairwise relations, (ii) representation of belief and attitude states as Ising model spins, $s \in \{-1, +1\}$, as opposed to binary variables, $s \in \{0, 1\}$, and (iii) the non-dependence of interacton effect magnitude on belief states.

Consider the following (hypothetical) motivating example. Suppose that parents' attitudes toward childhood vaccination typically depend, in a simplistic way, on their perceived relative risk of a vaccine, and the disease it protects against. Positive attitudes prevail when the disease is considered more risky or dangerous than the vaccine. Values around family wellbeing conceivably influence the relationship between perceived risk and vaccine attitudes, serving to amplify the existing effect.

Suppose that in an attempt to promote increased childhood vaccination rates, we undertake a media campaign appealing to family wellbeing values. The expected outcome is evident: attitudes improve for individuals who are relatively more concerned about the disease; for other individuals the opposite effect ensues.

This relational structure can be seen as a triplet-interaction extension to the belief system model used in the present study. In a belief system model with triplet interactions, the *effective* influence relations between pairs of beliefs or attitudes can vary in accordance with the specific belief state. Findings from Brewer et al. (2017) suggest that such interventions, which selectively leverage or amplify existing beliefs and attitudes may be more effective than interventions which attempt to change belief/attitude states. However, such relational structures cannot be captured in the pairwise model adopted here.

Next, we consider the implications of our decision to model belief and attitude states as spins, with values $s \in \{-1, +1\}$. Consider the following belief:

Believes that climate change is happening.

Which comprises both an epistemic position (*Believes that*) and a state-of-affairs (*climate change is happening*). Negating each of these components yields two reasonable choices for the 'opposite' state:

Does not *believe that climate change is happening.*

Believes that climate change is ***not*** *happening.*

As a consequence of our decision to model belief and attitude states as spins, we are assuming that beliefs and attitudes always emit influence on associated variables, regardless of their state. However, it is not clear that this assumption is always reasonable, in particular for beliefs where the opposite state is a negation of the epistemic position (the first example), or in the case of neutral or ambiguous states (van der Maas et al., 2026).

Finally, while the present model assumes that interaction effects in asymmetric belief systems are independent of the 'influencing' variable's state, we argue that there is reason to think that this may not always be true. Consider two variables:

- **Happening:** The belief that climate change is happening.
- **Action:** General attitude toward climate action.

An individual who *does not* believe in climate change logically should not support climate action. In the asymmetric belief system model, this corresponds to a large positive interaction, such that **Action** aligns with **Happening**. However, individuals who *do* believe in climate change may nonetheless oppose climate action for other reasons (e.g., cost or prioritisation), suggesting that **Action** may be less constrained when **Happening** is high than when it is low.

7.2 Implications for future work

Our findings suggest that while the symmetric assumption may often be valid, or at least a reasonable approximation, asymmetric relations between beliefs and attitudes are nonetheless possible. When true relations are asymmetric, not accounting for this when modelling belief interactions is a case of model misspecification, and can lead to incorrect inferences regarding the relative influence of different beliefs. The primary issue here is that symmetric models do not account for differences between a belief's influence (how much it affects the states of other beliefs) and influentiability (how much its own state is affected by other beliefs).

Despite the promising results of this study, several questions remain. The limited number of waves in the climate beliefs dataset prohibits us from distinguishing within-person and between-person effects, which is required to make strong claims regarding causal influences. We note that the broader climate attitudes survey does contain several additional waves, which are usable if we drop our requirements regarding the number of variables and inclusion of specific beliefs/attitudes. However, this presents a separate issue regarding the intervals between measurements, as inter-response times between different pairs of waves can differ significantly, violating the model requirements. Further consideration is required to determine whether, for instance, the additional waves can be used *only* to estimate individual baselines, while evenly-spaced observations are used to estimate interaction effects.

As discussed above, the regression decision tree approach to modelling the effect characterisation function can produce incomplete descriptions when important features are highly correlated (↩), and may not identify all effective-intervention conditions due to the somewhat arbitrary nature of tree depth to limit complexity (↩). The first issue can be resolved post-hoc (by assessing correlations with identified variables). One promising direction for the second is to identify more flexible rulesets, allowing arbitrary quantity and size, while penalising characterisation complexity directly, e.g., using description length (Aoga et al., 2018; Proença & van Leeuwen, 2020).

The models calibrated in Chapter 4 are assumed to reflect 'natural' belief system dynamics (i.e., minimal exogenous influence). By simulating interventions on these models, we are therefore assuming transferrability to situations *with* exogenous influences in the form of interventions. However, prior studies have demonstrated that belief system dynamics may differ in such situations, for instance due to increased salience of certain attitudes or beliefs (Unsworth & Fielding, 2014). As such, experimental validation—and, ideally, calibration to data collected in controlled intervention scenarios—is a natural continuation to the present study.

Our findings also suggest and support several broader directions for future research. Firstly, we posited two explanations for the asymmetric relations observed with regards to political attitudes and climate-related worry (↩). These are retrospectively applied to the findings, so arguably have minimal evidential weight (Popper, 1963). However, they demonstrate how the asymmetric non-equilibrium belief system model may be used to test (as opposed to generate) hypotheses about the general mechanisms by which asymmetric belief/attitude relations may occur.

Secondly, while belief system structure is typically expected to vary on an individual basis (Brandt & Morgan, 2022; Morgan & Wisneski, 2017), calibrating individual belief system models is generally considered infeasible due to the associated data requirements (Brandt, 2022). Moreover, due to the potential stability of belief dynamics, as observed in the present study, even substantial individual-level data is likely to reflect only a small set of possible belief states, limiting counterfactual analysis in individual models. However, our findings demonstrate structural similarities, as well as differences, across ideological groups. This suggests that a middle-ground approach, between individual-level and population-level networks may be effective in capturing both the ways in which belief systems are different, and the ways in which they are similar. This is akin to the notion of partial-pooling in multi-level Bayesian models, which assumes individuals in a group vary with respect to one another, but are constrained by common parameters (Gelman et al., 2013; Gelman & Hill, 2007; McElreath, 2020). Note that this is distinct from the estimation of individual *baseline activations* for the purpose of inferring within-person associations.

In our experiments we considered interventions as attempts to change the state of a belief or attitude directly. However, previous studies indicate that in certain situations, interventions targeting belief system *structure* may be more effective (Aggarwal & Hughes, 2026; Brewer et al., 2017). Rather than attempting to affect beliefs and attitudes directly, structural interventions can be seen as amplifying or changing the influence between beliefs. Although we have not considered such interventions in this study, the open source *Ising* Python package[19] released alongside this study already supports these, inviting future work on this topic.

Finally, we adopt a simplistic view of the interface between belief systems and the external world. This is most evident, for instance, in our treatment of interventions as acting directly and identically on individual beliefs/attitudes, and the assumption that current events (e.g., the 2020 US presidential election) have minimal impact on observed belief dynamics. In reality, some beliefs or attitudes may be easier or harder to affect directly than others. Communication of interventions or current events may be noisy (e.g., subject to interpretation), potentially impacting the magnitude, direction, and set of affected beliefs/attitudes, also on an individual basis. Clearly, a more realistic treatment of this interface is required for the proposed methods to be useful in the context of actual interventions. From a modelling perspective, Dalege et al. (2025) consider a related problem, namely the interface between belief systems belonging to different individuals, modelling peer influence as indirect, via social beliefs which may only be indicative of the true external state.

[19]https://github.com/henry-zwart/ising

Chapter 8

Conclusions

Most studies on belief system structure or dynamics acknowledge the likelihood that some beliefs or attitudes may be more causally influential than others. Despite this, current research deals, almost universally, in belief system models which assume symmetric influence between beliefs and/or attitudes. Theoretical models which capture asymmetric relations remain scarce, as do empirical studies on the existence of such asymmetry and its potential implications for belief system dynamics.

In this study we have addressed both theoretical and empirical shortcomings. We first introduced the Kinetic Belief System model (KBS) as a theory-based model of belief system dynamics based on the Causal Attitude Network model (Dalege et al., 2016), which: (i) *does not* assume symmetric influence and (ii) *does not* assume equilibrium dynamics. We then used the KBS model, calibrated to a longitudinal dataset comprising beliefs and attitudes about climate change in the US, to address the four research questions outined in the first chapter.

Our empirical investigation identified several clear asymmetric influence relations which appeared to be structured around a small set of highly influential attitudes (political views and climate-related worry). This suggests that (i) asymmetry may be the exception rather than the norm, and (ii) asymmetry may be better characterised in terms of the difference between a belief's influence and influentiability *in general*, than as a property of specific relations. Furthermore, our simulated intervention experiments demonstrated that asymmetry in a belief system can meaningfully affect intervention effectiveness, in some cases potentially changing conclusions about where to intervene. These effects were most pronounced when targeting or intervening on variables with several asymmetric influence relations.

Our secondary investigation examined how intervention effectiveness varies among individuals, and potential differences between liberal and conservative belief systems. In the first case, intervention effectiveness was found to depend predictably on individuals' pre-intervention belief states. Interestingly, *all* highly effective intervention scenarios featured a low initial level of climate-related worry, suggesting that attitudes which are both highly influential and highly influentiable may serve as effective indirect pathways for various interventions.

Second, despite relatively smaller sample sizes, conservative and liberal belief systems differed in sparsity and specific belief relations, while also displaying broad structural similarities. Both findings align with those of previous studies using symmetric networks (Lee et al., 2024). The similarities suggest that while belief systems likely vary between individuals, they also likely display common structural features (e.g., based on shared experiences or world-views). The observed differences in pairwise relations when stratifying by political ideology suggest the presence of higher-order relations between beliefs. This is to say that the existence, strength, or direction of influence between two beliefs may depend on the specific state of a third (in this case, political ideology).

As we draw to a close, our findings invite continued investigation in several respects. Firstly, though we have used the KBS model here to identify asymmetric influence, it is equally amenible to testing pre-existing hypotheses about beliefs or attitudes which are more (or less) influential than they are subject to influence. Secondly, in the previous chapter we discussed a hypothetical example in which higher-order interactions result in complex intervention dynamics. Though not currently captured by the KBS model, our comparison of the conservative and liberal belief systems suggests this may be a worthwhile extension. Relatedly, although we have only considered interventions targeting belief and attitude *states*, prior studies suggest that amplifying certain *associations* between pre-existing beliefs (rather than trying to change them directly) may also be effective (Aggarwal & Hughes, 2026; Brewer et al., 2017). As such, studies considering structural interventions targeting influence relations themselves are a natural progression to the intervention experiments described here.

Finally, while our empirical findings are promising, they would be considerably strengthened by investigation into the extent to which the inferred belief systems reflect within-person effects, as opposed to between-person effects. Disentangling these effects has received increased attention in recent years (Brandt, 2022; Brandt & Morgan, 2022), yet remains difficult without substantial individual-level data. Although not unique to the present study, we consider addressing this limitation critical to establishing the relevance of our empirical findings.

Returning to the primary topic of our investigation, our findings support the existence of asymmetric relations between beliefs and attitudes about climate change. Crucially, we have also demonstrated that this *matters*, both for studies concerning belief system structure, and those concerning belief system dynamics. When true relations are asymmetric, the use of symmetric models amounts to misspecification. This can cause incorrect assessments of relation existence and strength, which are frequently used to assess belief/attitude importance via node centrality indices. Moreover, this has immediate and obvious dynamic implications, since variables deemed non-influential in the symmetric model may simply be non-influentia*ble* in reality.

Chapter 9

Climate beliefs dataset

In this chapter we outline the context and construction of the dataset used for model calibration in Chapter 3. We will first describe the broader context and complexities of the dataset underlying this study, before outlining our data validation and cleaning methods in Chapter 9.2. Finally, in Chapter 9.3 we detail the construction of the targeted dataset used for model calibration (Chapter 3).

While many empirical studies on belief systems rely on cross-sectional data (Lee et al., 2025; Powell et al., 2023; van Noord et al., 2025), longitudinal data is necessary to capture changes in individuals' internal cognitive states over time. In this study we are fortunate to have had access to data from the **Longitudinal Panel of Perceptions About Climate Change and Covid** (**CCCV**), a representative longitudinal survey comprising six waves of responses from individuals residing in the United States, collected between April 2020 and March 2023 (cf. Constantino et al. (2022)). The assessed dimensions include general demographic information (e.g., age, gender, education, and financial status), beliefs, attitudes, and experiences relating to concurrently-salient topics such as COVID-19, climate change, or the 2020 US presidential election, and support for hypothetical policies. In addition to a wide-form table of survey response data, the dataset includes a codebook which specifies, for each survey item, the question text, occurrence in survey waves, and conditional display logic where applicable. At present, the codebook is limited to Waves 1—5, i.e., excluding the sixth (final) wave.

However, the dataset is not without complexities. Survey participation varies, with each participant responding to a (possibly non-contiguous) subset of waves. Figure 9.1 shows the number of participants who have responded to each combination of survey waves, with a minimum count of 1500. While each individual wave has a relatively high response rate (roughly 4000–5000), this rapidly drops off when other waves are

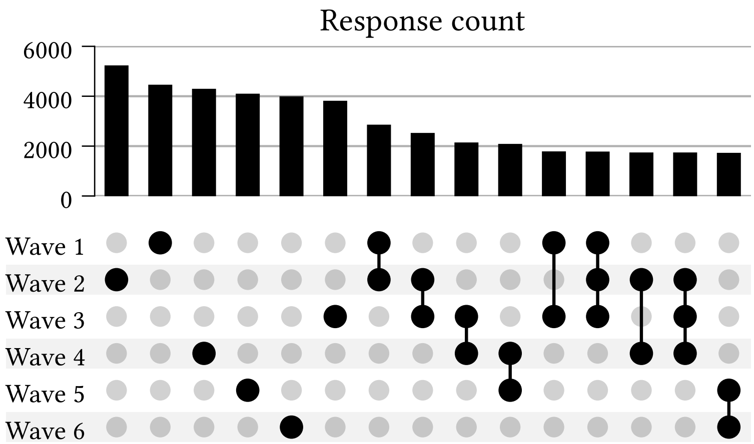


Figure 9.1: Number of repeat participants for different wave combinations in the Longitudinal Panel of Perceptions About Climate Change and Covid dataset, prior to dataset cleaning. Excludes combinations with fewer than 1500 responses.

considered. For instance, only ~2500 individuals responded to Waves 1 and 2, and only ~1900 of these individuals also responded to Wave 3.

Survey content also varies between waves and participants, both with regards to *which* questions are included, and *how* they are presented. Certain questions are displayed only in particular waves, or only to either repeating or new participants. Some questions are shown conditionally based on an individual's responses to prior questions (within the same wave). Others vary according to survey treatment conditions, such that individuals in different treatment groups are presented different variants of a question. This survey logic is not always executed correctly; in some cases participants are shown survey questions incorrectly (e.g., 'new' participants shown questions intended only for 'repeating' participants). We discuss this issue further in Chapter 9.1.

Question format, text, and response schemas also occasionally change between survey waves. Changes in response schema are less common, however, and typically affect items with categorical responses. With a view to constructing the calibration dataset for our experiments, we note that most response schemas are not binary and therefore require binarisation (see Chapter 3.2).

Casting our attention to the survey timing, we examine the survey response dates for each wave (Figure 9.2) and the distribution of interval durations between consecutive-wave responses across individuals (Figure 9.3), i.e., the 'inter-response time'.

In Figure 9.2 we observe that the survey waves occur with irregular spacing and duration. Some consecutive pairs of waves are much closer than others. For instance, notice that:

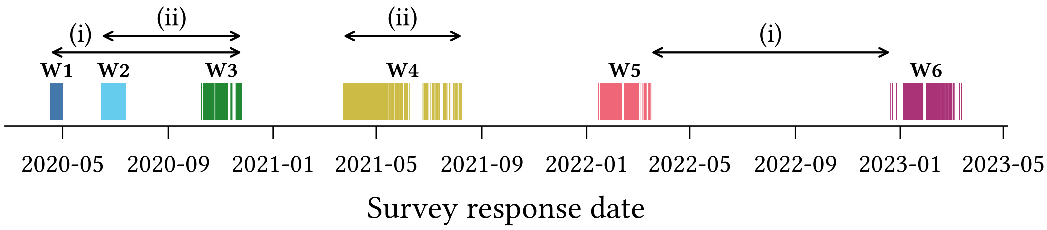


Figure 9.2: Per-wave survey response dates for the Longitudinal Panel of Perceptions About Climate Change and Covid dataset (prior to dataset cleaning). Annotations illustrate differences in intervals spanned by/between survey waves.

(i) The duration from the start of Wave 1 until the end of Wave 3 is shorter than the duration *between* Waves 5 and 6, and

(ii) The duration spanned by Wave 4 alone exceeds that of the interval spanned by both Waves 1 and 2.

This poses a potential problem for model calibration, since the Kinetic Belief System model (defined in Chapter 2) operates on the assumption that samples are equispaced.

We see the irregular spacing reflected also in the inter-response time distributions. For each pair of consecutive waves, we observe substantial variation in the time between responses for different participants.

Finally, we note that the time interval spanned by the longitudinal dataset includes several notable events which could reasonably be expected to influence — and confound — the dynamics of beliefs and attitudes in myriad contexts. These include the COVID-19 pandemic, which arrived in the US only three months prior to the first survey wave (Holshue et al., 2020), the 2020 US Presidential Election which occurred during Wave 3, and the January 6 United States Capitol Attack, which occurred between Waves 3 and 4.

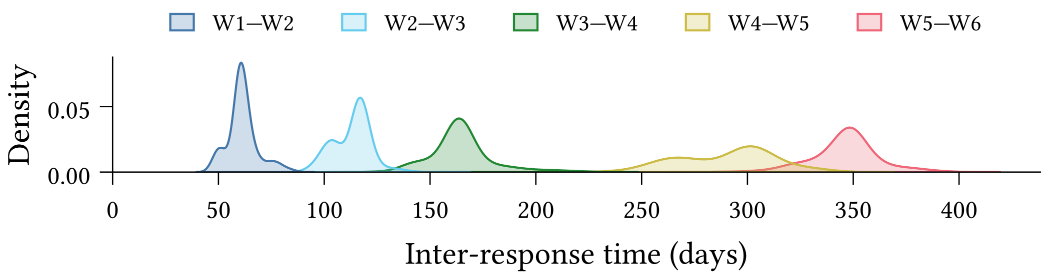


Figure 9.3: Inter-response time distribution between each pair of consecutive waves in the Longitudinal Panel of Perceptions About Climate Change and Covid dataset (prior to dataset cleaning). The inter-response time between two waves is calculated, for individuals present in both waves, as the number of days between their responses.

9.1 Data Validation

The complexity of the climate attitudes survey, as outlined above, necessitates a rigorous approach to data validation, both to identify errors in the expected schema and to ensure that the data matches our expectations.

We have implemented a general validation pipeline comprising three stages:

1. **Type-level validation:** The set of columns is exactly as expected, and all columns have the correct data type.

2. **Response-value validation:** All *non-null* responses are valid according to the schema specified in the codebook.

3. **Null-value validation:** Responses are null if, and only if, we expect them to be null.

9.1.1 Type-level validation

Type-level validation ensures that the *observed* data schema matches the *prescribed* schema. Since response data types vary between questions (see Table 9.1), the type-checking must be flexible and capable of handling complex data types.

For instance, categorical multiple-choice responses are represented using a `list[Enum]`, where `Enum` is a question-specific enum-type[20]. Type validation for a multiple-choice question thus requires checking: (i) that the response column comprises `list`'s, and (ii) that all list elements belong to the set of values defined by the `Enum`.

Response format	Raw type	Coerced type
Text	`str`	`str`
Single response (ordinal)	`int`	`int`
Single response (categorical)	`int`	`Enum`
Multiple response	`str`	`list[Enum]`
Numeric	`float`	`float | int`

Table 9.1: Data type coercion mapping for different question types in the Longitudinal Panel of Perceptions About Climate Change and Covid survey.

[20]An enum is a type defined by a finite set of allowable values. In our case the values are human-readable strings. For instance, the `dem_urban` survey question, which asks 'What kind of area do you live in?' has responses with data type described by the enum $\{\text{Urban}, \text{Suburban}, \text{Rural}\}$.

9.1.2 Response-value validation

Response-value validation then ensures that all non-null response values are valid according to the survey codebook. For most variables this is straightforward. Numeric and single-response ordinal variables typically have a defined range (e.g., $18 \leq \text{age} \leq 99$, or $1 \leq x \leq 5$ for a 5-point Likert scale variable). Text-entry responses are always considered valid.

Single-response categorical questions have `Enum` type, so type-level validation is sufficient to ensure that the responses do not contain any values outside the set defined by the `Enum`. However, the allowable values occasionally change between survey waves. Hence response-value validation is also required for categorical single-response and multiple-response variables, and in general must handle schema variation between waves.

9.1.3 Null-value validation

Finally, null-value validation ensures that responses are null if, and only if, they are expected to be null. For a question Q, the response of a participant P in wave W is permitted to be null, iff, at least one of the following four conditions is true:

N1. Q is not asked in wave W,

N2. In wave W, Q is only shown to *new* participants, but P is a *repeating* participant (or vice versa),

N3. In wave W, Q is only shown to a treatment group of which P is not a member, or

N4. In wave W, Q is conditionally shown to participants based on their response to a distinct question Q', and P's response to Q' does not satisfy the condition.

Formally we can write the condition for P's response R being null as:

$$\text{null}(R) \iff (\text{N1} \lor \text{N2} \lor \text{N3} \lor \text{N4}) \tag{9.1}$$

Null-value validation then consists in identifying the set of responses $\boldsymbol{R}$ such that for $R \in \boldsymbol{R}$, exactly one of $\text{null}(R)$ and $(\text{N1} \lor \text{N2} \lor \text{N3} \lor \text{N4})$ is true. Validation succeeds if $\boldsymbol{R}$ is empty. Otherwise the members of $\boldsymbol{R}$ are counterexamples to the above conditions.

9.1.4 Implementation

We implement type-level and simple response-value validation using the Pandera Python library for DataFrame validation (Bantilan et al., 2022). We implement manual validation checks for more complex response-value cases, such as questions whose response schema varies between waves.

The null-value validation logic, i.e., the process of testing for contradictions Equation 9.1, is also manually implemented. Conditions N1 and N2 are derived automatically from the survey codebook. Conditions N3 and N4 must be manually specified, as the codebook currently does not have a standardised method for describing conditional display logic.

9.1.5 Validation results

At the time of writing, approximately 25% of the complete survey question set for Waves 1–5 has been validated, including all survey questions which were considered for the targeted calibration dataset (Chapter 9.3). Since Wave 6 is currently undocumented in the survey codebook we can reasonably validate neither the data schema, nor the presence of null-values. We have therefore not yet validated any of the data from Wave 6, and exclude this wave from the present study.

All type-level and response-value validation checks succeed, providing a strong guarantee that the data schema matches our expectations per the codebook. We do, however, encounter several problems during null-value validation. Sara Constantino's guidance was instrumental in the process of diagnosing these problems.

In some cases, these were due to errors in the codebook itself. These errors are relatively straightforward to identify from the null-value validation results, since they often affect all individuals in a particular wave. For instance, if the codebook specified that a question is not presented in Wave 1, yet all responses are non-null, then this most likely indicates an error in the codebook. While some cases are more subtle, such as where a treatment class is misspecified, when arising due to a codebook error we still expect the contradictions to affect a well-defined subset of the population (in this case, the individuals in a particular treatment class).

In other cases we identify contradictions which are due to errors in the survey process, which caused certain survey questions to be displayed to some individuals in conditions which did not satisfy the specified survey logic. Figure 9.4 shows examples observed in Waves 2 and 3. Some cases were systematic, affecting all individuals until the survey process was updated (*left*); however, in other cases we have not been able to identify the

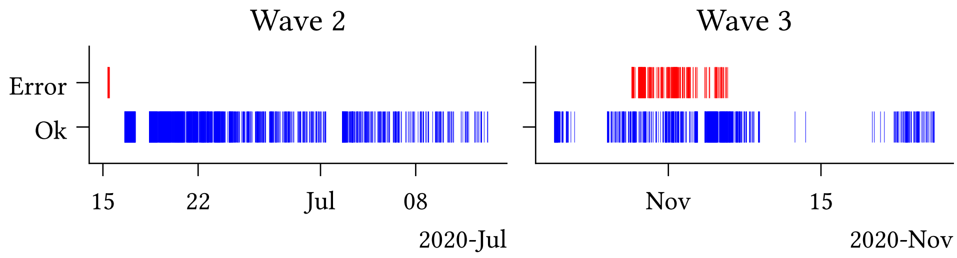


Figure 9.4: Systematic (*left*) and unresolved (*right*) null-value validation errors identified in Waves 2 and 3 of the Longitudinal Panel of Perceptions About Climate Change and Covid survey.

source of the error (*right*). We exclude all responses which fail the null-value validation from further analysis in the data cleaning stage (Chapter 9.2.2).

The described validation process is primarily concerned with ensuring that the data schema — comprising types and values — aligns with our expectations as per the survey codebook. While comprehensive in this regard, the process does not account for *all* possible forms of errors or inconsistencies. One category which is currently unaccounted for, but is worth mentioning, comprises inconsistencies in the responses from a given individual over multiple waves. For instance, the dataset includes several cases in which individuals presented the question:

"Were you born in the United States?"

respond with 'Yes' in one wave, but 'No' in another. Validating data for inconsistencies such as this requires careful consideration on a question-by-question basis to assess response types for conflicts. We consider this category beyond the scope of this study, describing it here only to illustrate the bounds of the above validation process.

9.2 Normalisation, cleaning, and transformations

We apply pre-processing steps to make the dataset amenible to downstream analysis, which are divided into three categories:

- **Normalisation:** Ensuring the data schema (variable names, types, and values) conform to consistent standards.
- **Cleaning:** Filtering erroneous or otherwise-problematic data.
- **Transformation:** Structural dataset enrichments, including constructed data variables and improved organisation of information.

9.2.1 Normalisation

Variable names in the original dataset are subject to some inconsistency, featuring a mixture of `snake_case`, `CamelCase`, `dot.case`, and `kebab-case`. We therefore first convert the names of all variables specified in the codebook to a standard format, making all names lowercase, and replacing hyphens and periods with underscores.

The dataset also contains several columns which are not described in the codebook, often containing survey metadata, the functions of which are typically indicated using a variable name prefix. For instance, the `Group_` prefix indicates that a column describes treatment group membership. We do not normalise these variable names here, so as to maintain identiability for later transformations (Chapter 9.2.3).

Secondly, we coerce all variable data types to a standard set. In most cases this is straightforward. Dates and times represented as strings are coerced to timezone-free `datetime` types. Non-floating-point numeric values (which constitute most survey response types) are coerced to 64-bit integers.

Categorical variable responses are originally represented as integeres, which presents three issues. Firstly, it imposes an ordinal structure which does not, in general, make sense for categorical variables. For instance, consider the first four response options to a survey question asking participants about their recent experience with different forms of extreme weather:

1. Severe winter storm, 2. Hurricane, 3. Tornado, 4. Heat wave

One may devise any number of reasonable ordinal interpretations over this set (e.g., ‘highest average risk of property damage’, or ‘annual emergency response cost’), yet such interpretations are context-dependent and not inherent to the categories themselves. Recording categorical responses using numeric types therefore risks spurious interpretations. Second, integer values provide no information about the categories to consumers of the dataset, requiring that they cross-reference manually with the codebook. Thirdly, in some cases the mapping from categories to integers for a given question varies between survey waves. We account for all three of these issues by coercing categorical variables to `Enum` types, which preserve information about the underlying categories, and do not assume an ordinal structure.

Multiple-selection survey items are more complex still. These are represented as strings comprising comma-separated integers, whose values typically refer to non-ordinal categories. We coerce these to `list[Enum]` types, which have all the benefits of the categorical enums discussed above, and enable simpler programmatic analysis (e.g., testing for set membership, determining differences in response sets between waves).

Finally, we perform a series of value updates to ensure consistency between responses to different questions. This consists in replacing all empty string responses with `null` values[21], and re-mapping integer columns (both numeric and ordinal) such that the minimum value is zero.

9.2.2 Cleaning

Following schema normalisation, we then perform a cleaning stage, which primarily includes filtering out invalid or problematic data. We only filter out entire survey responses, rather than answers to specific survey items. Survey responses may be removed for any one of three reasons.

Firstly, we remove responses where the `participant_id` column is `null`, such that the individuals cannot be traced across survey waves. Secondly, we remove any responses from participants who fail the survey validity check in *any* of their participation waves. Thirdly, we remove any responses which fail either of the response-value or null-value validation checks described in the previous section (Chapter 9.1). We recognise that these issues are not necessarily problematic in every research context (e.g., cross-sectional studies are unconcerned with tracing an individual's responses across waves), and thus have made each condition optional in the dataset construction code.

9.2.3 Transformation

Finally, we apply several transformation steps with the goal of simplifying manipulation and analysis of the existing dataset, or enriching it with additional information.

The original dataset includes two survey items querying participants' alignment with the two largest US political parties (i.e., the Republicans and the Democrats). The first item asks whether the participant whether they consider themselves as *Republican*, *Democrat*, or *Independent*. Those who respond *Independent* are then presented a follow-up question asking whether they *lean* toward either (or neither) party. We replace these two survey items with a single variable combining the responses, with possible values:

$$\text{Democrat} \prec \text{Leaning Democrat} \prec \text{Independent} \prec \text{Leaning Republican} \prec \text{Republican}$$

[21]Note that this particular value mapping must happen *after* null-value validation (Chapter 9.1).

While this supplements the first item with additional information, we note two potential limitations of this transformation. Firstly, some participants who respond *Republican* or *Democrat* to the first question may more accurately describe themselves as only leaning in a given direction, if they had been aware of the (conditional) second question. Therefore the ‘Leaning’ responses are perhaps better interpreted as ‘Independent with Republican/Democrat tendencies’.

Secondly, our implicit assumption that *Independent* is a midpoint between *Republican* and *Democrat* would misrepresent individuals who consider themselves ‘right-of-Republican’ or ‘left-of-Democrat’. However, the available data does not allow us to distinguish such cases. Furthermore, this issue is also present in the original first question.

Our second transformation concerns survey questions with different variants, which are conditionally displayed based on treatment group membership. The original dataset includes, for each treatment group associated with a given question, a separate response column, as well as a binary indicator column describing group membership. Since group membership is always mutually exclusive in the dataset, we coalesce these various columns, replacing them with singular response and group membership columns. The group membership column contains integer indexes as opposed to the original binary indicators. In some cases, different treatment groups have an associated numerical value. For instance, the `ccSolveX` items query support policies compensating communities affected by climate change, with the treatment group specifying the cost (X) of such a policy to the participant. In these cases we supplement the dataset with an additional column containing the associated numerical value, simplifying downstream analyses.

Finally, we improve the dataset organisation by constructing a database comprising tables for survey item and question metadata and responses, as well as participants themselves (Figure 9.5). The requirement for such as database arose during the construction of the climate beliefs dataset used for model calibration (Chapter 9.3). The original data, comprising a codebook (a spreadsheet) and an RData file containing rows of participant responses, is not immediately amenible to our task, which requires reasoning about both survey logic and participation in tandem. The constructed database primarily serves to link survey metadata (extracted from the codebook) with survey response data.

Recognising the hierarchical structure of the longitudinal dataset, in which a single survey item may vary in question text or response schema between waves or treatment conditions, we separate the concepts of *survey items* (the underlying concept being assessed) and *survey questions* (the particular mode of assessment). Survey questions are associated with particular waves, participant types (i.e., new or repeating), and treatment conditions, but may be related under a common survey item. *Survey responses*

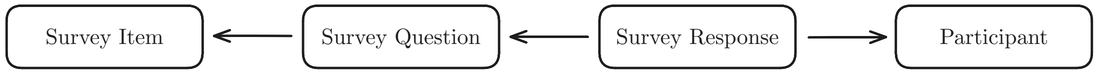


Figure 9.5: The constructed survey database comprises tables for survey items (underlying concepts assessed in the survey), survey questions (the particular mode in which an item is assessed in a given wave or treatment condition), survey responses, and participants. Arrows denote data hierarchy, e.g., each response belongs to a participant, and is associated with a particular survey question.

are associated with a particular question as well as a *survey participant*. We include a separate table for participants which records wave participation as well as the initial wave in which a participant has responded.

The database is stored on disk as a collection of parquet files, which preserve complex data type information, including those in Table 9.1, and are supported in both R (e.g., using Arrow) and Python (e.g., using Pandas or Polars).

9.3 Climate beliefs dataset

In light of the discussed complexities and breadth of content of the CCCV survey, we construct a smaller, targeted dataset of beliefs and attitudes relating to climate change, which we expect—on theoretical grounds—to exhibit interdependent behaviour. This serves as the calibration dataset for the experiments described in this study. We will refer to this targeted dataset as the **climate beliefs dataset**.

We aim for 7–10 variables in the final set, to keep both the number of model parameters and parameter uncertainty acceptably small. We first filter the survey items to consider only those which assess cognitive states (e.g., beliefs, attitudes, opinions, stances), and which either relate directly to climate change, or which are expected to influence climate-related items (e.g., political alignment). The parameter estimation method (outlined in **TODO**) imposes several additional requirements on the constructed dataset, in particular, the dataset should comprise at least two waves, with approximately equispaced observations per-individual, and with no null values. We additionally constrain the dataset to include three specific variables of interest:

1. **CC Worry**: Level of worry regarding current and future climate change.

2. **CC Others Worry**: Belief about how worried *others* are about climate change.

3. **CC Human**: Belief that climate change is caused by human activities.[22]

Climate-related worry is generally considered an influential factor for other attitudes relating to climate change, including support for climate policy (Bouman et al., 2020; Bumann, 2021; Goldberg et al., 2021; Mead et al., 2012; Whitmarsh et al., 2022), and is expected to be influenced by beliefs others' level of worry, as a descriptive norm (Gavrilets et al., 2024). The third variable serves to provide a more complex view of individuals' beliefs about the nature of climate change; while most Americans believe in the existence of climate change, there is greater variation in beliefs about its causes (Hamilton et al., 2015).

To satisfy these requirements while maximising the number of observations (for model calibration purposes), we limit the climate beliefs dataset to waves 3 and 4, which contain 1693 repeat observations, excluding survey errors. After removing items with no substantial correlations, and small groups with no substantial external correlations, we identify 17 relevant survey items, comprising both beliefs and attitudes (see Appendix C for the full set of items).

However, this set of items still exceeds our target range of 7–10. We therefore proceed to identify groups of similar or redundant variables which may be removed or combined into interpretable index variables. Note that cross-sectional methods should not be used for this analysis, as they fail to capture temporal relatonships, which are fundamental to the kinetic belief system model (defined in Chapter 2). Instead, we examine the temporal and contemporaneous networks obtained using lag-1 vector autoregression (VAR). These networks are derived from the solution to the following regression problem, described in Epskamp, Waldorp, et al. (2018):

$$\begin{aligned} \boldsymbol{y}^t &= \boldsymbol{B}\boldsymbol{y}^{t-1} + \varepsilon^t \\ \varepsilon^t &\sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Theta}) \end{aligned} \tag{9.2}$$

Where $\boldsymbol{y}^t$ is the vector of observed variable values for each individual at time t. The *temporal network* matrix $\boldsymbol{B}$ comprises linear next-state predictors, while the *contemporaneous network* matrix $\boldsymbol{K} = \Theta^{-1}$ describes the pairwise conditional correlations which remain after accounting for temporal effects. The entries of $\boldsymbol{K}$ are analogous to partial correlations, constituting the conditional correlation between X and Y given the remaining variables.

[22]In the CCCV survey, the item `CC Human` actually has four possible (categorical) responses, reflecting all combinations of 'human activities' and 'natural causes' as the causes of climate change. We remap these values to 'human-caused' and 'not human-caused', such that the variable is instead binary, and thus both amenible to the analysis described below and interpretable in the kinetic belief system model in Chapter 4.

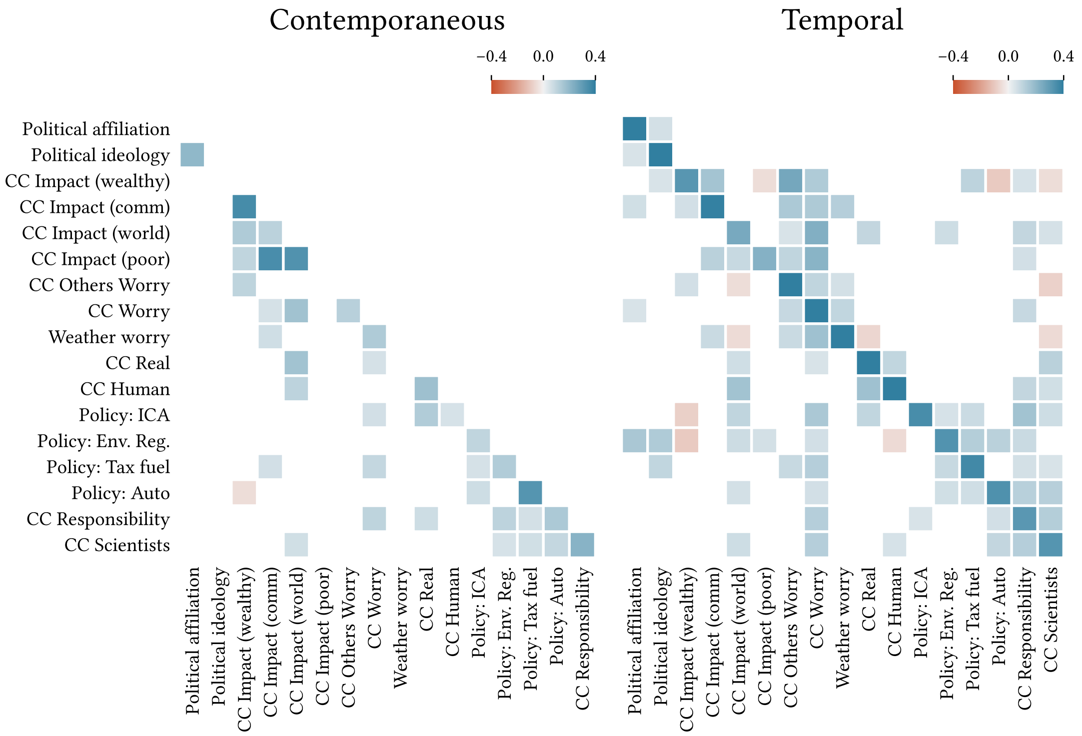


Figure 9.6: Contemporaneous (left) and temporal (right) network matrices for the filtered candidate set. Entries in row i of the temporal matrix are the regression predictors for the subsequent value of variable X_i. Rows and columns are ordered using hierarchical clustering on the temporal matrix: $d_{\boldsymbol{K}}(X,Y) = 1 - |\boldsymbol{K}|$.

Figure 9.6 displays the contemporaneous and temporal networks for the filtered set of 17 candidate variables[23], with rows and columns ordered using hierarchical clustering on the temporal matrix to highlight groups of related items. Values with magnitude less than 0.05 are excluded, as is the (symmetric) upper triangle of the contemporaneous network. The values in each row of the temporal matrix are the regression predictors for that row's variable. Therefore we interpret row i as describing the variables which *influence* item i, and column i as describing the variables which are influenced *by* item i.

We note three apparent clusters of variables: (i) political views, (ii) beliefs about climate change impacts, and (iii) positions on climate policy, and other attitudes relating to climate action. In the first case, political affiliation (partisan identity) and ideology (conservative ⟷ liberal) cluster together in both networks, yet appear to have minimal external associations. We retain this group on theoretical grounds, since political attitudes are often key determinants for climate-related attitudes and policy positions (Bumann, 2021; Palm et al., 2017). Secondly, beliefs regarding the impact of climate change on different population groups, `CC Impact (X)`, exhibit strong internal contemporaneous associations, relatively weaker temporal associations. We choose here

[23]We have re-coded the survey items such that the matrix values are predominantly positive.

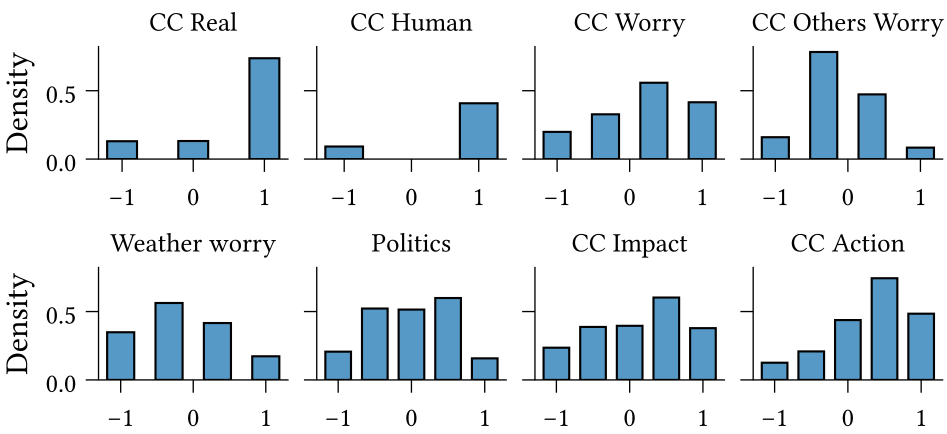


Figure 9.7: Marginal distribution for each of the eight variables in the climate beliefs dataset (Table 9.2).

to combine these into an index variable, although an alternative strategy would be to retain only `CC Impact (World)` and/or `CC Impact (Wealthy)`, noting that these variables each have several (and distinct) temporal associations. Finally, the right-most six variables form clear clusters in both networks. Four of these relate directly to climate policy, while the remaining two assess participants views on the importance of individual action in managing climate change, and the role of scientists in guiding the climate change response. We combine these into a third cluster, which we interpret as reflecting an individual's general attitude toward climate action.

We construct the index variables by first re-scaling each constituent item to the interval $[-1, 1]$ and then taking the average. All other variables are also re-scaled in the same way. The final dataset comprises eight variables, described in Table 9.2. Figures 9.7 and Figure 9.8 show the marginal distributions, and the contemporaneous and temporal networks, respectively, for the climate beliefs dataset.

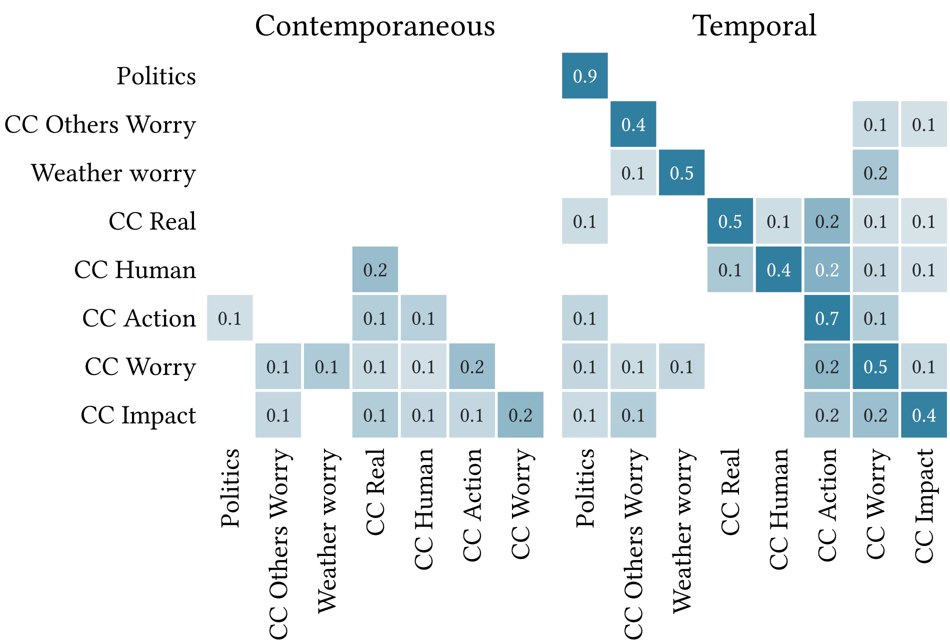


Figure 9.8: Contemporaneous and temporal network matrices for the climate beliefs dataset. Entries in row i of the temporal matrix are the regression predictors for the subsequent value of variable X_i. Rows and columns are ordered using hierarchical clustering on the temporal matrix: $d_{\boldsymbol{K}}(X, Y) = 1 - |\boldsymbol{K}|$.

Item	Index	Interpretation
CC Real	No	Belief that climate change is/is not real.
CC Human	No	Belief that climate is/is not caused (at least in-part) by human activities.
CC Worry	No	Level of worry about current and future climate change.
CC Others Worry	No	Belief regarding *others'* level of worry about current and future climate change.
Weather Worry	No	Level of worry about possible near-term extreme weather events or natural disasters.
Politics	Yes	Political views, as a combination of partisan alignment and political ideology.
CC Impact	Yes	Belief regarding the degree of current climate change impacts, in general.
CC Action	Yes	General attitude toward action on climate change.

Table 9.2: Variables included in the climate beliefs dataset. Index variables are constructed by taking the average of their constituent columns, after re-scaling to the interval $[-1, 1]$.

Chapter 10

Ethics and Data Management

A new requirement for the thesis is that there must be a short section in which you reflect on the ethical aspects of your project. This requirement is related to one of the final objectives that a graduated student of the Master of Computational Science must meet: “The graduate of the program has insight into the social significance of Computational Science and the responsibilities of experts in this field within science and in society“. You don’t need to devote an entire chapter to this; a short section or paragraph is sufficient.

I acknowledge that the thesis adheres to the ethical code (https://student.uva.nl/en/topics/ethics-in-research) and research data management policies (https://rdm.uva.nl/en) of UvA and IvI.

The following table lists the data used in this thesis (including source codes). I confirm that the list is complete and the listed data are sufficient to reproduce the results of the thesis. If a prohibitive non-disclosure agreement is in effect at the time of submission “NDA” is written under “Availability” and “License” for the concerned data items.

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Appendix A

Derivations

A.1 Model

A.1.1 Positive edges implying consistency possible

Part 1:

1. Internal consistency is possible, iff, for a given model, there exists a configuration of belief states such that every pair of positively associated beliefs have the same state, and every pair of negatively associated beliefs have opposite states.

2. Corrolary: Internal consistency is not possible, iff, for every configuration, there exist a pair of beliefs which are in conflict. i.e., where positively associated beliefs have opposing states, or negatively associated beliefs have the same state.

3. Fit model to dataset

4. Signs of edges denote whether each pair of beliefs is aligned or misaligned under the original variable coding

5. Reversing the scale for a variable has the effect of multiplying its edge signs by -1.

6. Notice that there is a bijective mapping between possible re-codings of the dataset, and possible model configurations. The result follows immediately.
    1. Any re-coding which makes all edges positive is an example of internal consistency in the original model.
    2. If for every re-coding, there exists at least one negative edge, then internal consistency is impossible.

7. Therefore, internal consistency is possible, if and only if, it is possible to re-code the dataset variables such that there are no negative edges in the resulting model.

A.2 Parameter estimation

A.2.1 Log-likelihood

Equation 3.5

A.2.2 Expected log-likelihood

Equation 3.6

A.2.3 Partial derivatives of expected log-likelihood

A.3 Dataset

A.3.1 Binarisation probability

Equation 3.1

Appendix B

Additional results

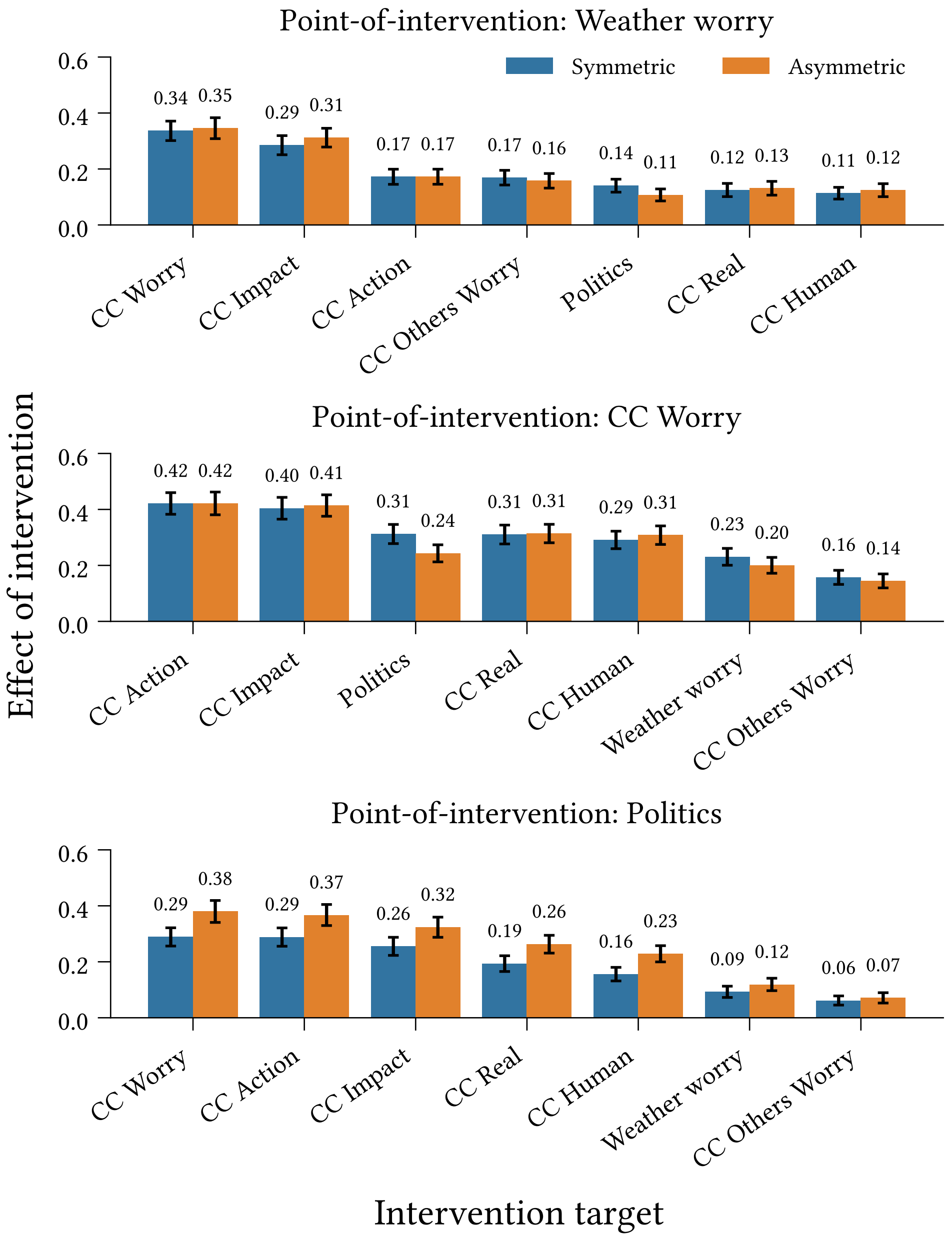

Figure B.1: Outbound effect of intervention (Definition 5.2.1) for interventions targeting `Weather Worry`, `CC Worry`, and `Politics`, in symmetric and asymmetric belief system models calibrated to the climate beliefs dataset. Measurements taken at $t = 10$ (approximately five years in the models' timescale). Intervention strength: $\delta_h = 2.5$. Confidence intervals display 1.96 standard deviations around the mean effect, measured across 500 repeated simulations.

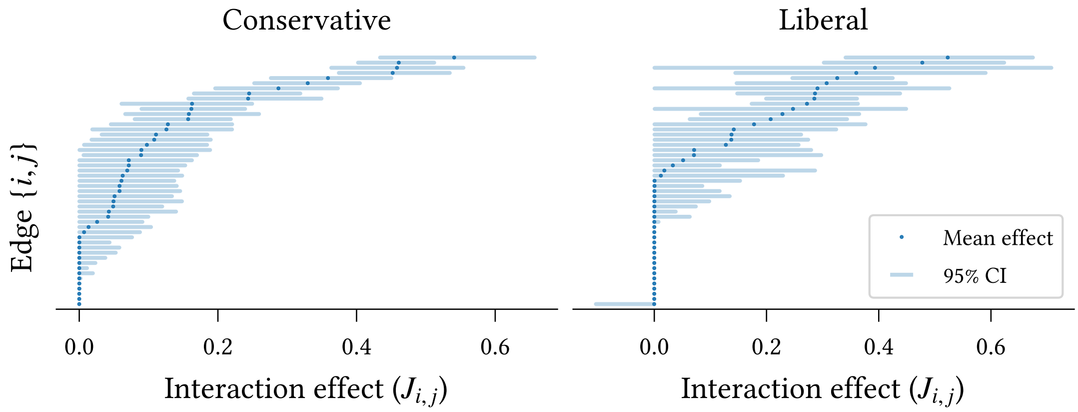

Figure B.2: Edge accuracy in ideology-specific models.

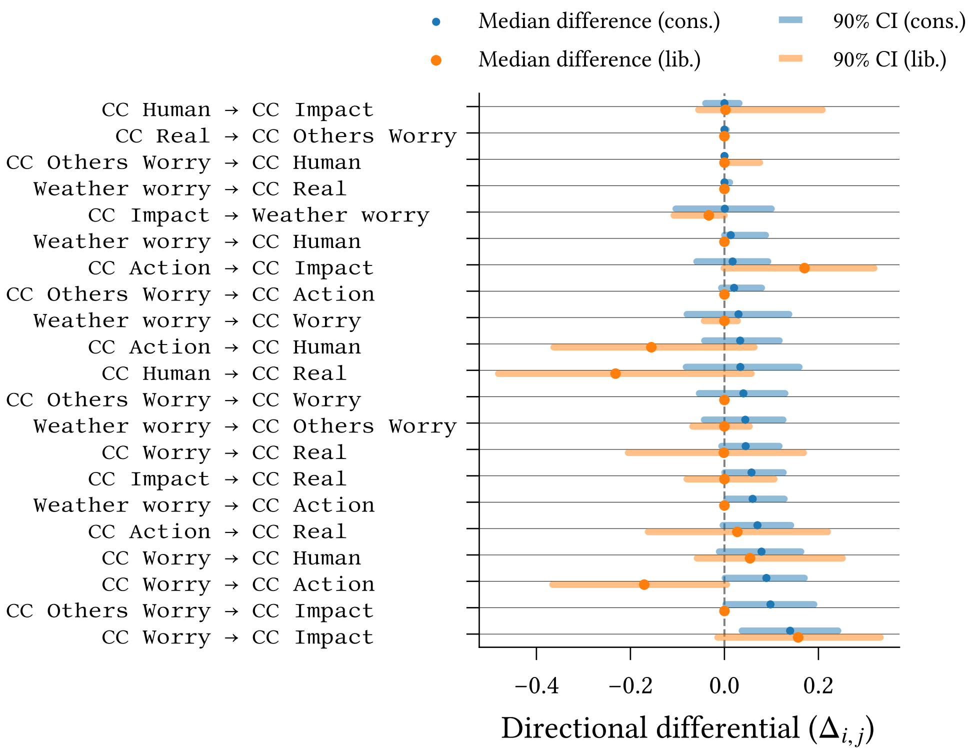

Figure B.3: Directional differentials for ideology-specific-models.

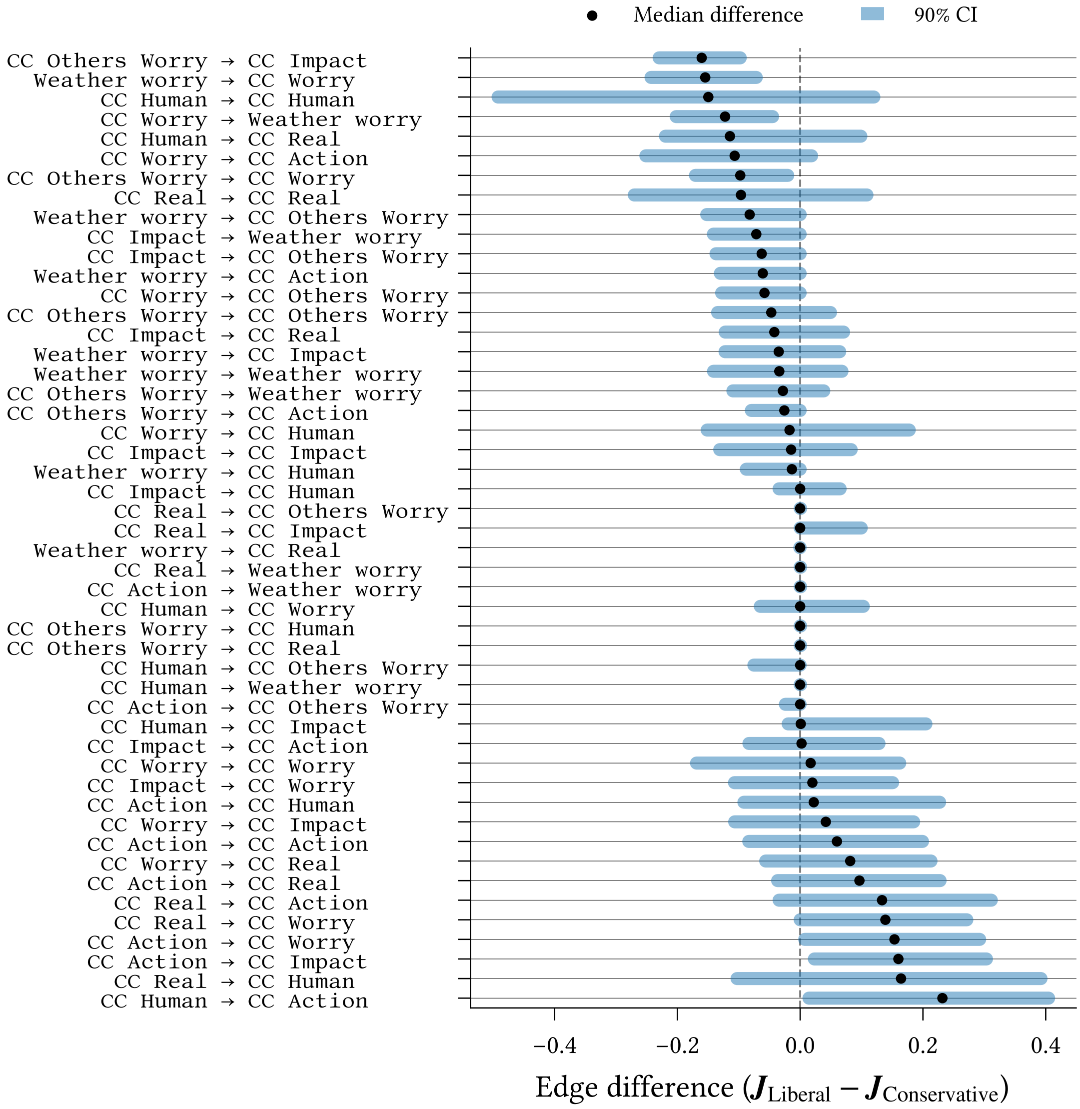

Figure B.4: Differences in interaction effect between ideological models.

Appendix C

Climate Beliefs Dataset

Item	Question text
CC Real	Do you think that climate change is happening?
CC Human	Do you think rising temperatures are a result of human activities, natural causes, or both?
CC Impact X	How much do you think climate change is currently harming X in general? Where X is one of: "the world", "wealthy communities in the United States", "poor communities in the United States", "your local community"
CC Others Worry	How worried do you think most Americans are about global warming/climate change these days?
CC Worry	How worried are you about current and future global warming/climate change?
Weather Worry	How worried are you about an extreme weather event or natural disaster happening to you personally in the next year?
CC Responsibility	It is important that individuals take action on issues of climate change.
CC Scientists	Scientists with appropriate expertise should guide how we respond to climate change
Policy: ICA (*)	Do you favour an international agreement committing the USA and other countries to reduce their carbon emissions?
Policy: Tax fuel	How much do you support/oppose a tax on the production/distribution of carbon-based fuelds?
Policy: Auto	How much do you support/oppose stronger carbon emissions standard for car manufacturers?
Policy: Env. Reg.	Which statement comes closer to your views?
Political affiliation (**)	In politics today, do you consider yourself a:
Political ideology	What is your political ideology?

Table C.1: Complete set of filtered survey items used to construct the climate beliefs dataset (Chapter 9.3)